DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

WA-554

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	554	5FA	500A-WA-FR1-N-4501-01	1 OF 1	1	CJ-M15W14-PR-A20-000167	2A	
2	554	4FA	400A-WA-FR1-N-4501-01	1 OF 1	0	CJ-M15W14-PR-A20-000167	2A	
3	554	3FA	400A-WA-FR1-N-4501-01	1 OF 1	0	CJ-M15W14-PR-A20-000167	2A	
4	554	2FA	400A-WA-FR1-N-4501-01	1 OF 1	0	CJ-M15W14-PR-A20-000167	2A	
5	554	2FA	300A-WA-FR1-N-4529-01	1 OF 2	0	CJ-M15W14-PR-A20-000167	2A	
6	554	2FA	300A-WA-FR1-N-4529-01	2 OF 2	0	CJ-M15W14-PR-A20-000167	2A	
7	554	2FA	250A-WA-FR1-N-4648-01	2 OF 2	0	CJ-M15W14-PR-A20-000167	2A	
8	554	5FA	350A-WA-FR1-N-4527-01	1 OF 1	1	CJ-M15W14-PR-A20-000167	2A	
9	554	RFA	350A-WA-FR1-N-4527-01	1 OF 2	0	CJ-M15W14-PR-A20-000167	2A	
10	554	RFA	350A-WA-FR1-N-4527-02	2 OF 2	0	CJ-M15W14-PR-A20-000167	2A	
11	554	RFA	200A-WA-FR1-N-4532-01	1 OF 1	0	CJ-M15W14-PR-A20-000109	1A	
12	554	RFA	250A-WA-FR1-N-4533-01	1 OF 1	0	CJ-M15W14-PR-A20-000108	1A	
13	554	4FA	80A-WA-FR1-N-4528-01	1 OF 1	0	CJ-M15W14-PR-A20-000167	2A	
14	554	4FA	50A-WA-FR1-N-4535-01	1 OF 1	0	CJ-M15W14-PR-A20-000151	1A	
15	554	4FA	50A-WA-FR1-N-4536-01	1 OF 1	0	CJ-M15W14-PR-A20-000152	2A	
16	554	3FA	80A-WA-FR1-N-4655-01	1 OF 1	0	CJ-M15W14-PR-A20-000143	2A	
17	554	RFA	200A-WA-FR1-N-4531-01	1 OF 1	0	CJ-M15W14-PR-A20-000107	1A	
18	554	RFA	100A-WA-FR1-N-4651-01	1 OF 1	0	CJ-M15W14-PR-A20-000142	1A	
19	554	RFA	250A-WA-FR1-N-4534-01	1 OF 1	0	CJ-M15W14-PR-A20-000110	1A	

DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

WA-554

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
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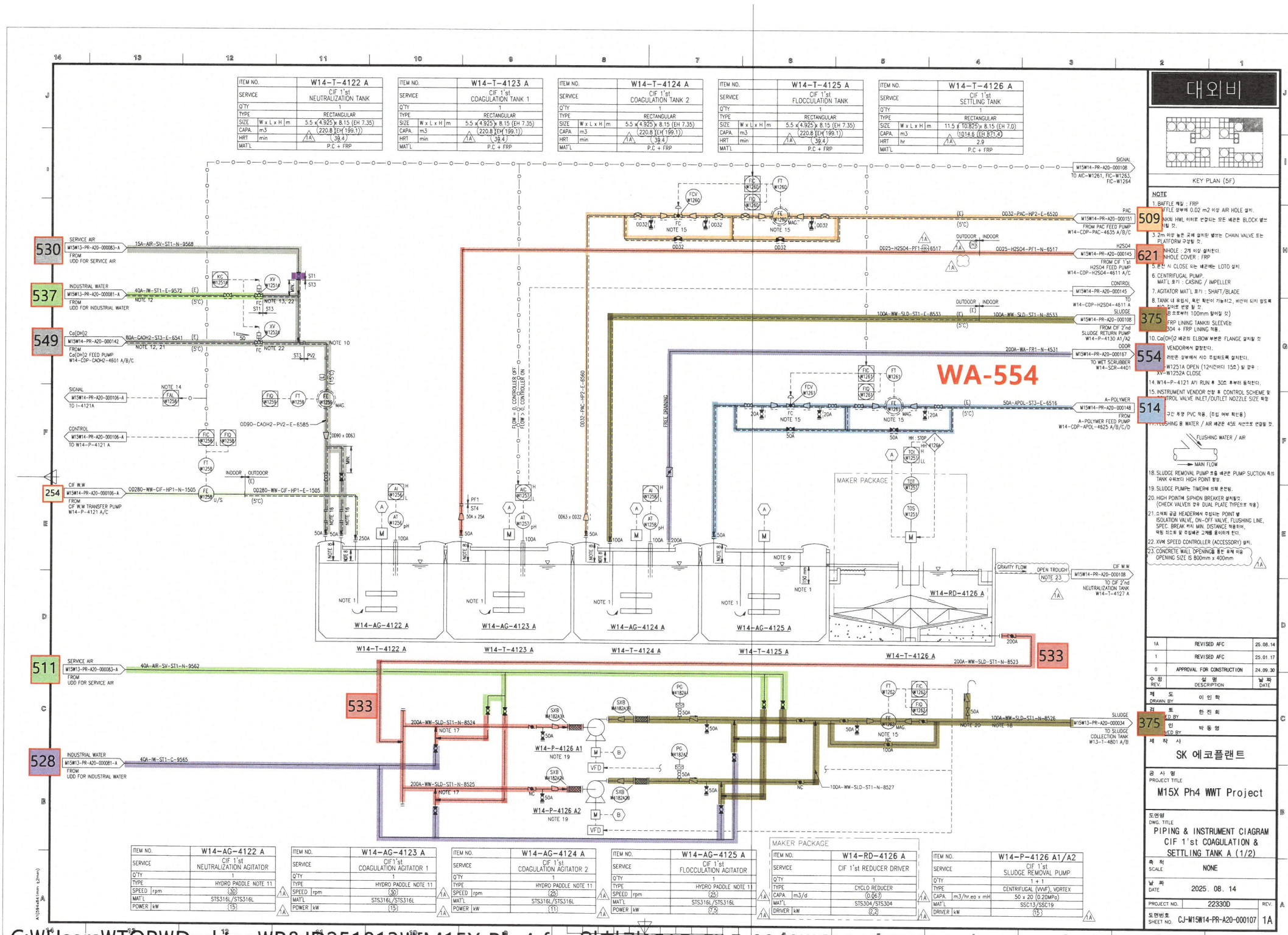
21	554	2FA	250A-WA-FR1-N-4650-01	1 OF 1	0	CJ-M15W14-PR-A20-000136	1A	
22	554	2FA	80A-WA-FR1-N-4649-01	1 OF 1	0	CJ-M15W14-PR-A20-000136	1A	
23	554	2FA	80A-WA-FR1-N-4647-01	1 OF 1	0	CJ-M15W14-PR-A20-000135	1A	
24	554	2FA	100A-WA-FR1-N-4584-01	1 OF 1	0	CJ-M15W14-PR-A20-000163	1A	
25	554	2FA	000A-WA-FR1-N-7006-01	1 OF 1	0	CJ-M15W14-PR-A20-000167	2A	


대외비
KEY PLAN (5F)
WA-554
W14-T-4122 A
CIF 1'st NEUTRALIZATION TANK
W14-T-4123 A
CIF 1'st COAGULATION TANK 1
W14-T-4124 A
CIF 1'st COAGULATION TANK 2
W14-T-4125 A
CIF 1'st FLOCCULATION TANK
W14-T-4126 A
CIF 1'st SETTLING TANK
W14-AG-4122 A
W14-AG-4123 A
W14-AG-4124 A
W14-AG-4125 A
W14-RD-4126 A
W14-P-4126 A1
W14-P-4126 A2
MAKER PACKAGE
NOTE
1. BAFFLE 재질 : FRP
2. BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
3. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치 것.
4. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성될 것.
5. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
6. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
7. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
8. AGITATOR MAT'L 표기 : SHAFT/BLADE
9. TANK 내 유입시, 육안 확인이 가능하고, 바닥이 더지 않도록 유입부로부터 100mm 떨어질 것.
10. Ca(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
11. VENDOR에서 결정한다.
12. 라인은 상부에서 시수 주입하도록 설치한다.
13. XV-W1251A OPEN (12시간마다 15초) 및 경우 : XV-W1252A CLOSE
14. W14-P-4121 A/I RUN 후 30초 후부터 동작한다.
15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
16. 구간 제외 PVC 적용. (진입 여부 확인필)
17. FLUSHING 용 WATER / AIR 배관은 45도 사선으로 연결할 것.
18. SLUDGE REMOVAL PUMP 흡입 배관은 PUMP SUCTION 측의 TANK 수위보다 HIGH POINT 형성.
19. SLUDGE PUMP는 TIMER에 의해 운전됨.
20. HIGH POINT에 SIPHON BREAKER 설치할것. (CHECK VALVE의 경우 DUAL PLATE TYPE으로 적용)
21. 소석회 공급 HEADER에서 주입되는 POINT 별 ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE, SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 막힘 최소화 및 주입배관 교체를 용이하게 한다.
22. XVM SPEED CONTROLLER (ACCESSORY) 설치.
23. CONCRETE WALL OPENING을 통한 유체 이송 OPENING SIZE IS 800mm x 400mm
1A REVISED AFC 25.08.14
1 REVISED AFC 25.01.17
0 APPROVAL FOR CONSTRUCTION 24.09.30
수정 REV. 설명 DESCRIPTION 날짜 DATE
제도 DRAWN BY 이 인 학
검토 CHKD BY 한 전 희
승인 APVED BY 박 동 영
제작사
SK 에코플랜트
공사명 PROJECT TITLE
M15X Ph4 WWT Project
도면명 DWG. TITLE
PIPING & INSTRUMENT CIAGRAM
CIF 1'st COAGULATION & SETTLING TANK A (1/2)
축척 SCALE NONE
날짜 DATE 2025. 08. 14
PROJECT NO. 22330D
REV. 1A
도면번호 SHEET NO. CJ-M15W14-PR-A20-000107
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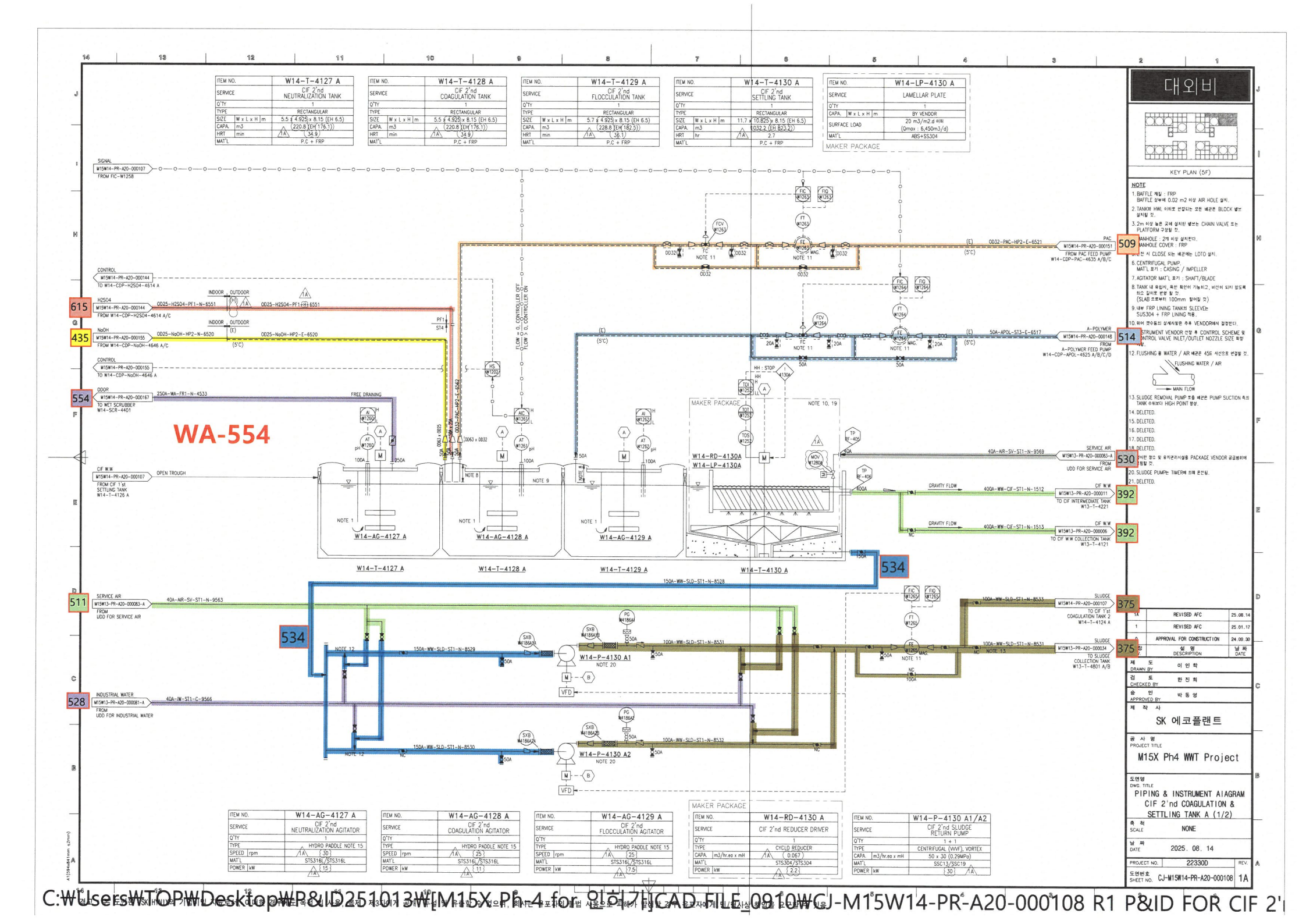

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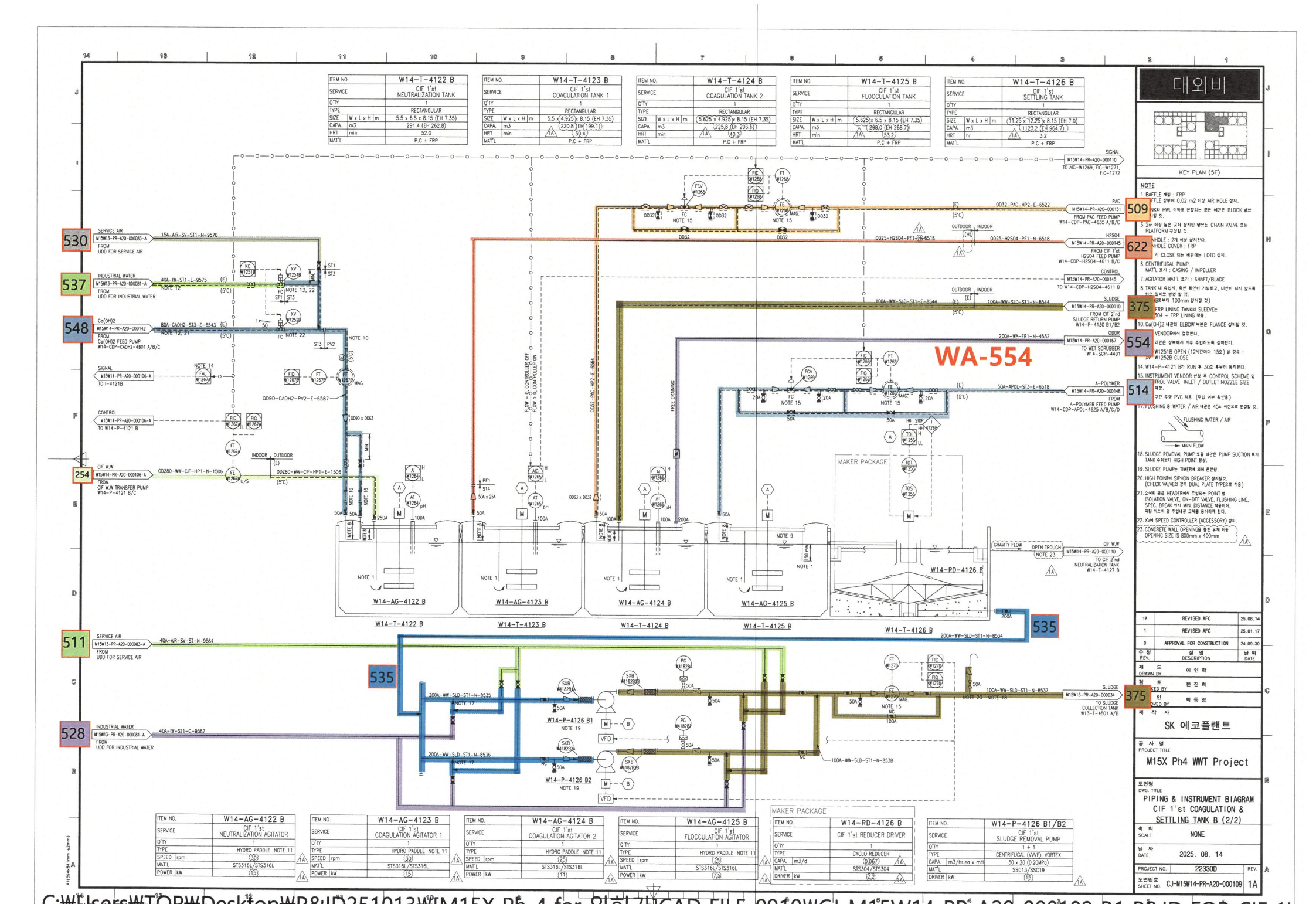

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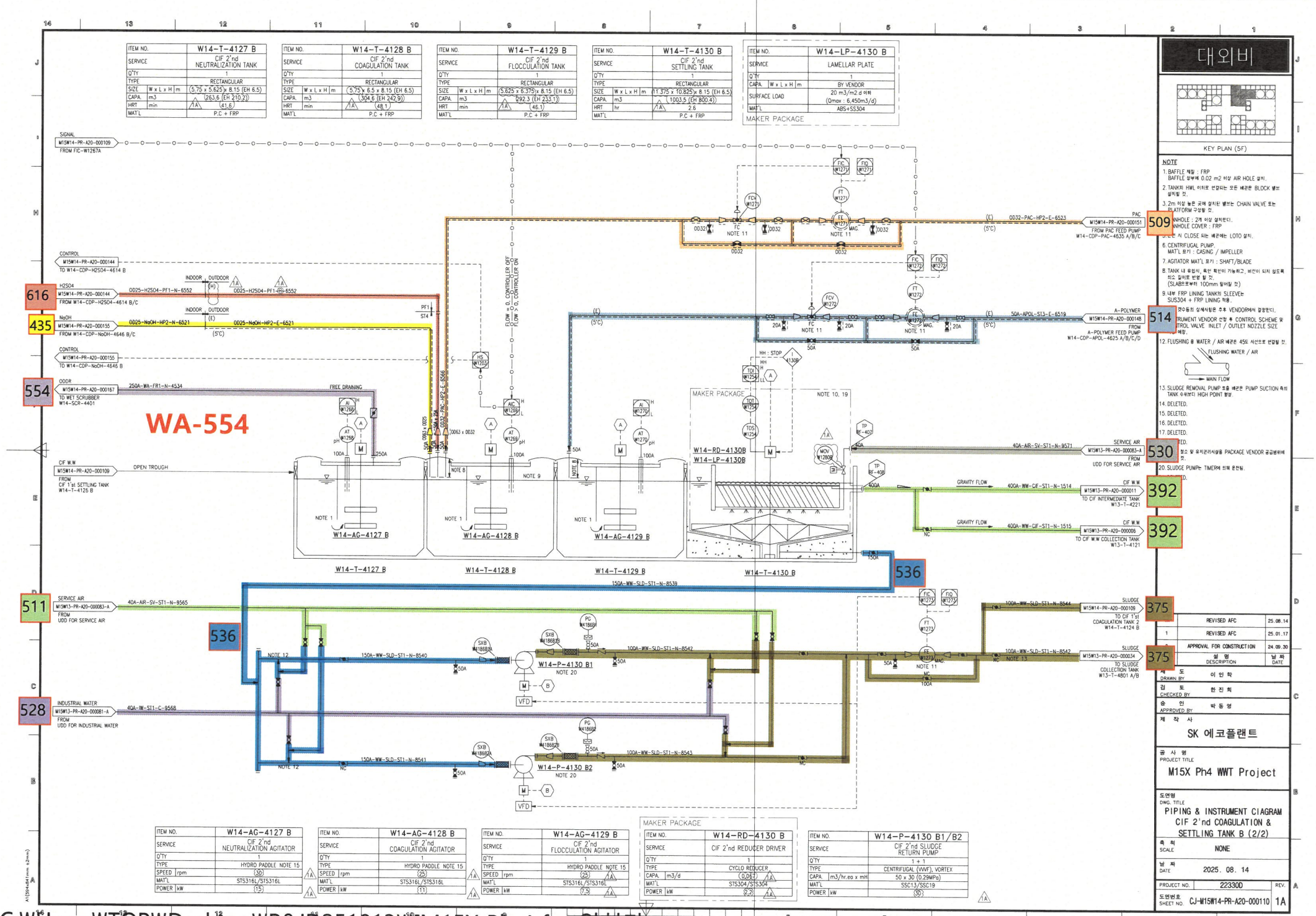

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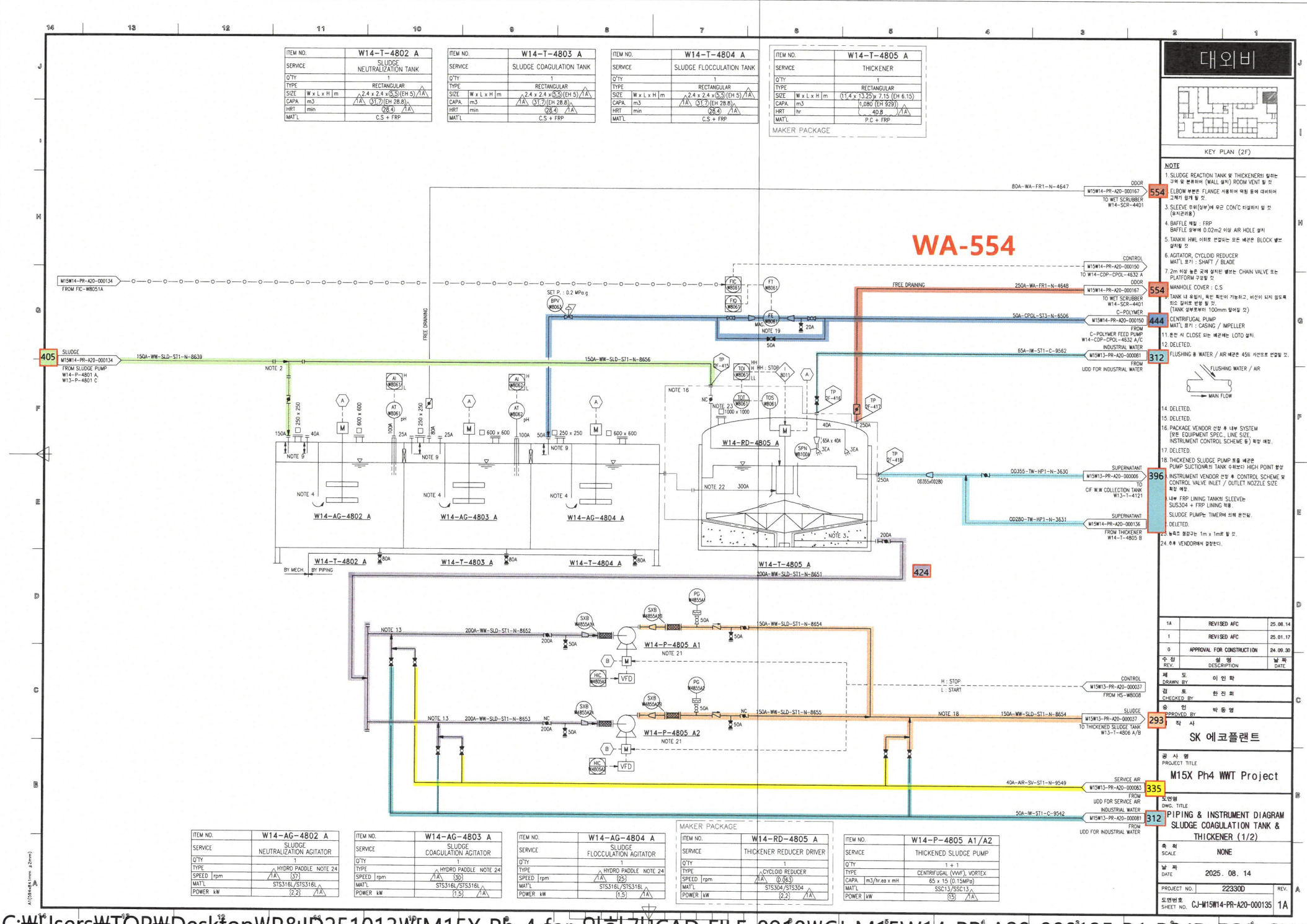
대외비
WA-554
W14-T-4802 A
SLUDGE NEUTRALIZATION TANK
W14-T-4803 A
SLUDGE COAGULATION TANK
W14-T-4804 A
SLUDGE FLOCCULATION TANK
W14-T-4805 A
THICKENER
MAKER PACKAGE
W14-RD-4805 A
W14-P-4805 A1
W14-P-4805 A2
W14-AG-4802 A
W14-AG-4803 A
W14-AG-4804 A
NOTE
1. SLUDGE REACTION TANK 및 THICKENER의 일반는 구역 및 분류하여 (WALL 설치) ROOM VENT 할 것.
2. ELBOW 부분은 FLANGE 사용하여 역팀 등에 대비하여 고정시 쉽게 할 것.
3. SLEEVE 주위(상부)에 무근 CON'C 타설하지 말 것 (유지관리용)
4. BAFFLE 재질 : FRP
BAFFLE 상부에 0.02m2 이상 AIR HOLE 설치
5. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 벨브 설치할 것
6. AGITATOR, CYCLOID REDUCER
MAT'L 표기 : SHAFT / BLADE
7. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것
8. MANHOLE COVER : C.S
9. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 감쉬로 반영 할 것.
(TANK 상부로부터 100mm 밑에일 것)
10. CENTRIFUGAL PUMP
MAT'L 표기 : CASING / IMPELLER
11. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
12. DELETED.
13. FLUSHING 용 WATER / AIR 배관은 45도 사선으로 연결할 것.
14. DELETED.
15. DELETED.
16. PACKAGE VENDOR 선정 후 내부 SYSTEM (모든 EQUIPMENT SPEC., LINE SIZE, INSTRUMENT CONTROL SCHEME 등) 확정 예정.
17. DELETED.
18. THICKENED SLUDGE PUMP 토출 배관은 PUMP SUCTION부의 TANK 수위보다 HIGH POINT 형성
19. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 확정 예정
20. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
21. SLUDGE PUMP는 TIMER에 의해 운전됨.
22. DELETED.
23. 농축조 점검구는 1m x 1m로 할 것.
24. 추후 VENDOR에서 검토한다.
KEY PLAN (2F)
1A REVISED AFC 25.08.14
1 REVISED AFC 25.01.17
0 APPROVAL FOR CONSTRUCTION 24.09.30
수정 REV. 설명 DESCRIPTION 날짜 DATE
제도 DRAWN BY 이인학
검토 CHECKED BY 한진희
승인 APPROVED BY 박동영
작사
SK 에코플랜트
공사명 PROJECT TITLE
M15X Ph4 WWT Project
도면명 DWG. TITLE
PIPING & INSTRUMENT DIAGRAM SLUDGE COAGULATION TANK & THICKENER (1/2)
축척 SCALE NONE
날짜 DATE 2025. 08. 14
PROJECT NO. 22330D REV.
도면번호 SHEET NO. CJ-M15W14-PR-A20-000135 1A
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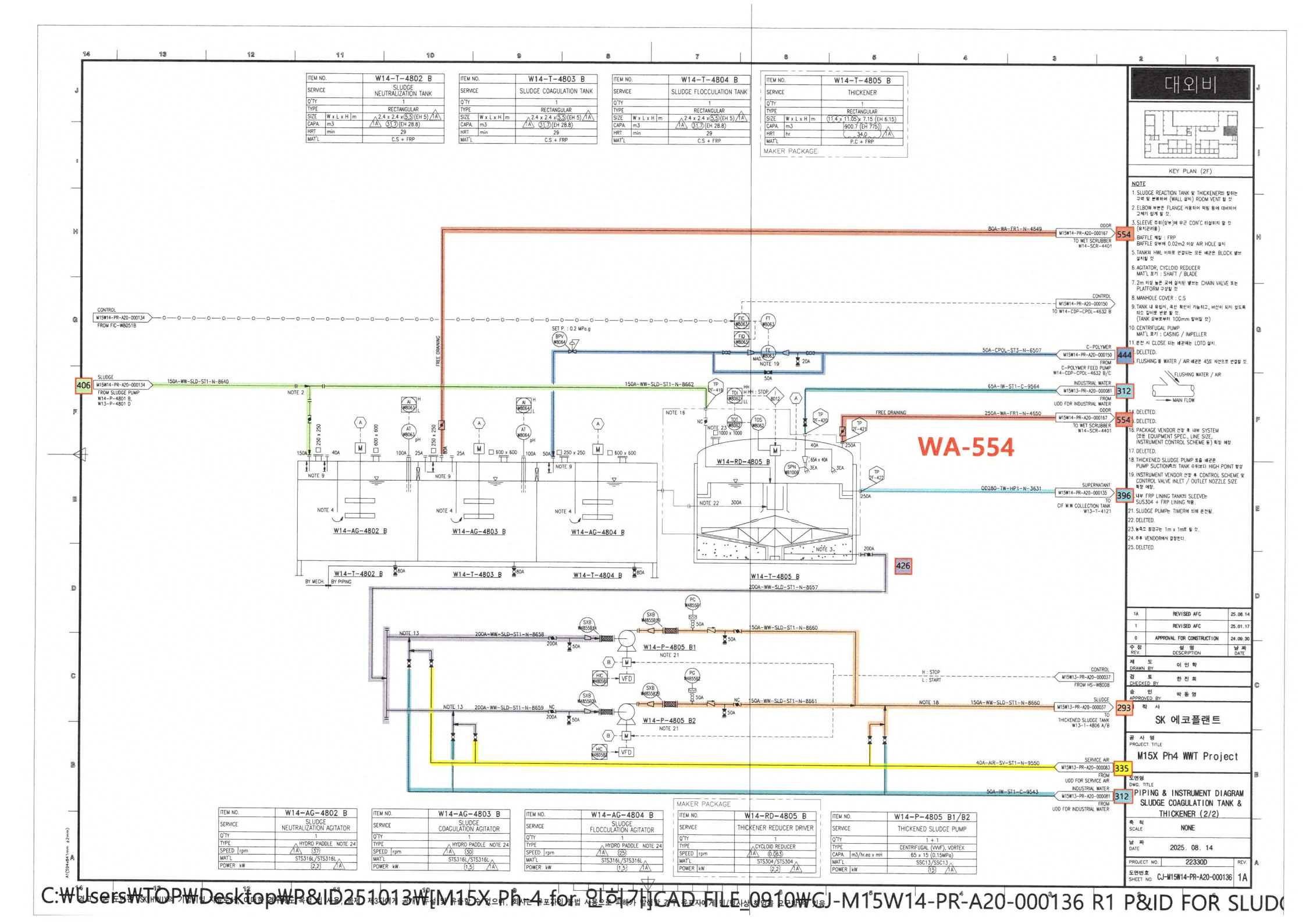
대외비
KEY PLAN (2F)
WA-554
W14-T-4802 B
SLUDGE NEUTRALIZATION TANK
W14-T-4803 B
SLUDGE COAGULATION TANK
W14-T-4804 B
SLUDGE FLOCCULATION TANK
W14-T-4805 B
THICKENER
MAKER PACKAGE
W14-RD-4805 B
W14-AG-4802 B
W14-AG-4803 B
W14-AG-4804 B
W14-P-4805 B1
W14-P-4805 B2
NOTE
1. SLUDGE REACTION TANK 및 THICKENER가 있는 구역 및 분류하여 (WALL 설치) ROOM VENT 할 것
2. ELBOW 부분은 FLANGE 사용하여 막힘 등에 대비하여 교체가 쉽게 할 것.
3. SLEEVE 주위(상부)에 무근 CON'C 타설하지 말 것 (유지관리용)
4. BAFFLE 재질 : FRP
BAFFLE 상부에 0.02m2 이상 AIR HOLE 설치
5. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것
6. AGITATOR, CYCLOID REDUCER
MAT'L 표기 : SHAFT / BLADE
7. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것
8. MANHOLE COVER : C.S
9. TANK 내 유입시, 육안 확인이 가능하고, 바인이 되지 않도록 하는 갑이로 분분 할 것.
(TANK 상부로부터 100mm 상이할 것)
10. CENTRIFUGAL PUMP
MAT'L 표기 : CASING / IMPELLER
11. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
12. DELETED.
13. FLUSHING 용 WATER / AIR 배관은 45도 사선으로 연결할 것.
14. DELETED.
15. DELETED.
16. PACKAGE VENDOR 선정 후 내부 SYSTEM (모든 EQUIPMENT SPEC., LINE SIZE, INSTRUMENT CONTROL SCHEME 등) 확정 예정.
17. DELETED.
18. THICKENED SLUDGE PUMP 토출 배관은 PUMP SUCTION부의 TANK 수위보다 HIGH POINT 형성
19. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 확정 예정.
20. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
21. SLUDGE PUMP는 TIMER에 의해 운전됨.
22. DELETED.
23. 농축조 점검구는 1m x 1m로 할 것.
24. 추후 VENDOR에서 검토한다.
25. DELETED.
1A REVISED AFC 25.08.14
1 REVISED AFC 25.01.17
0 APPROVAL FOR CONSTRUCTION 24.09.30
수정 REV. 설명 DESCRIPTION 날짜 DATE
제도 DRAWN BY 이인학
검토 CHECKED BY 한진희
승인 APPROVED BY 박동영
작사 SK 에코플랜트
공사명 PROJECT TITLE M15X Ph4 WWT Project
도면명 DWG. TITLE P&ID PIPING & INSTRUMENT DIAGRAM SLUDGE COAGULATION TANK & THICKENER (2/2)
축척 SCALE NONE
날짜 DATE 2025. 08. 14
PROJECT NO. 22330D REV.
도면번호 SHEET NO. CJ-M15W14-PR-A20-000136 1A
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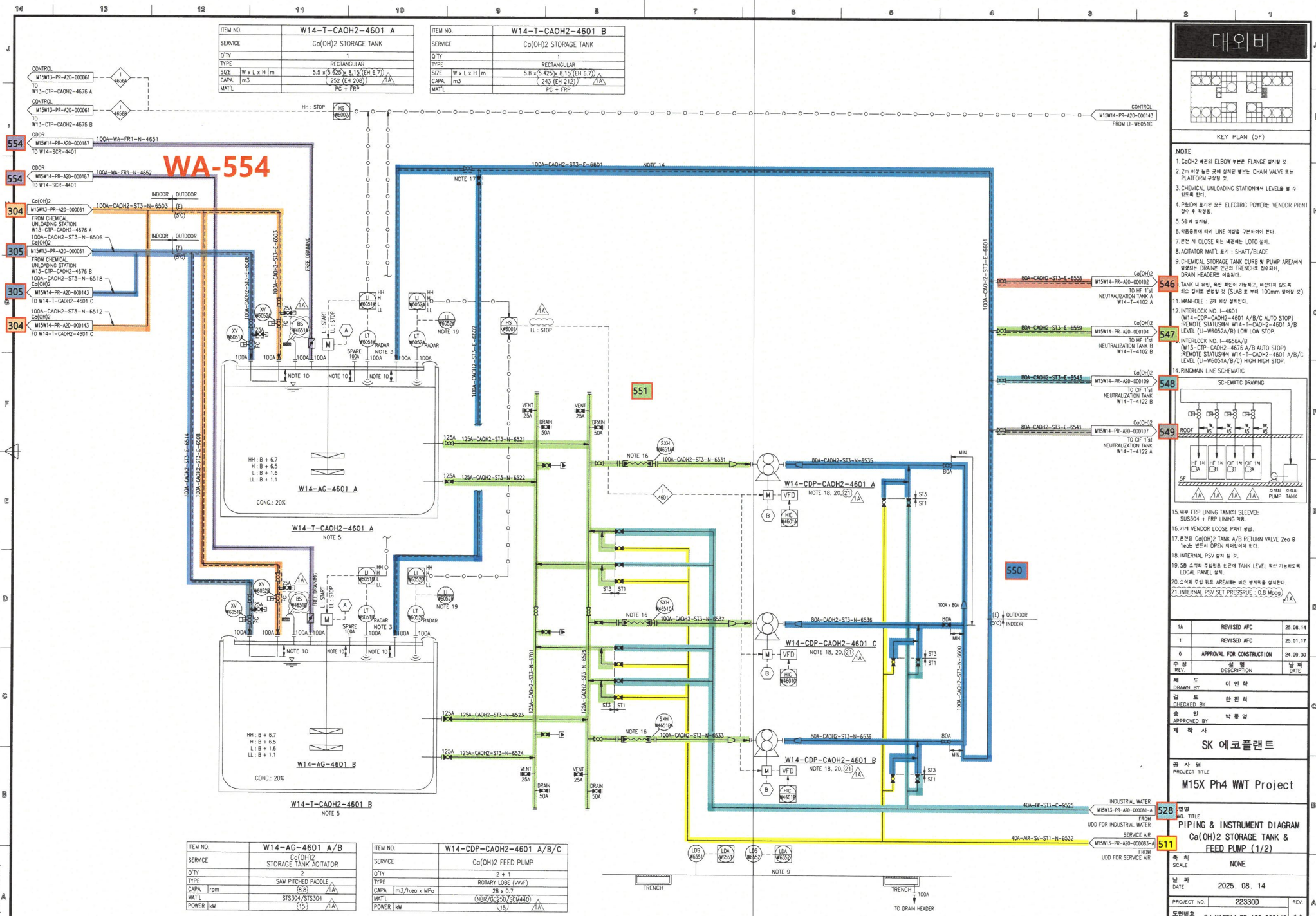
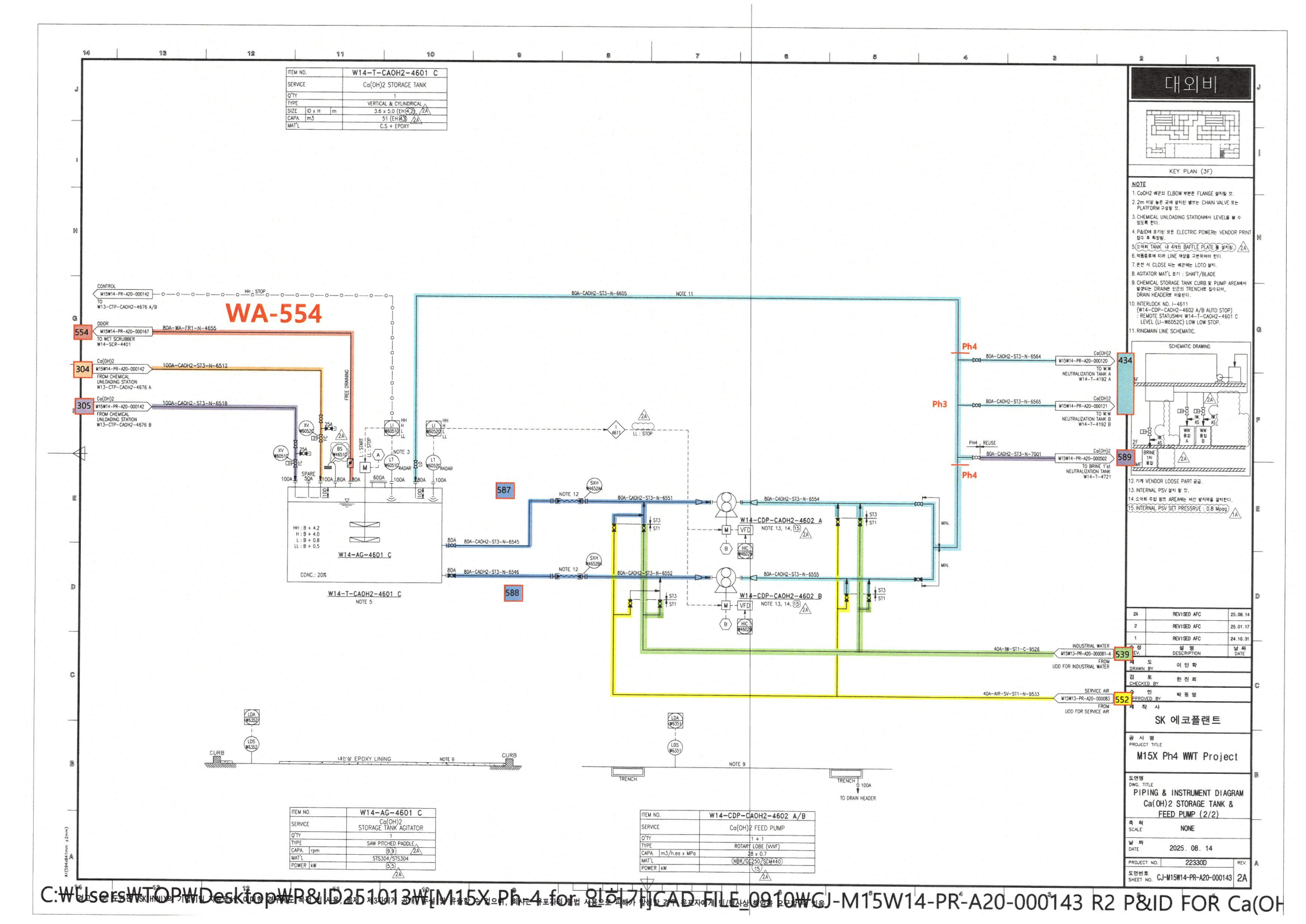


경고 : 본 도면은 SK 에코플랜트의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 인해서 발생한 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



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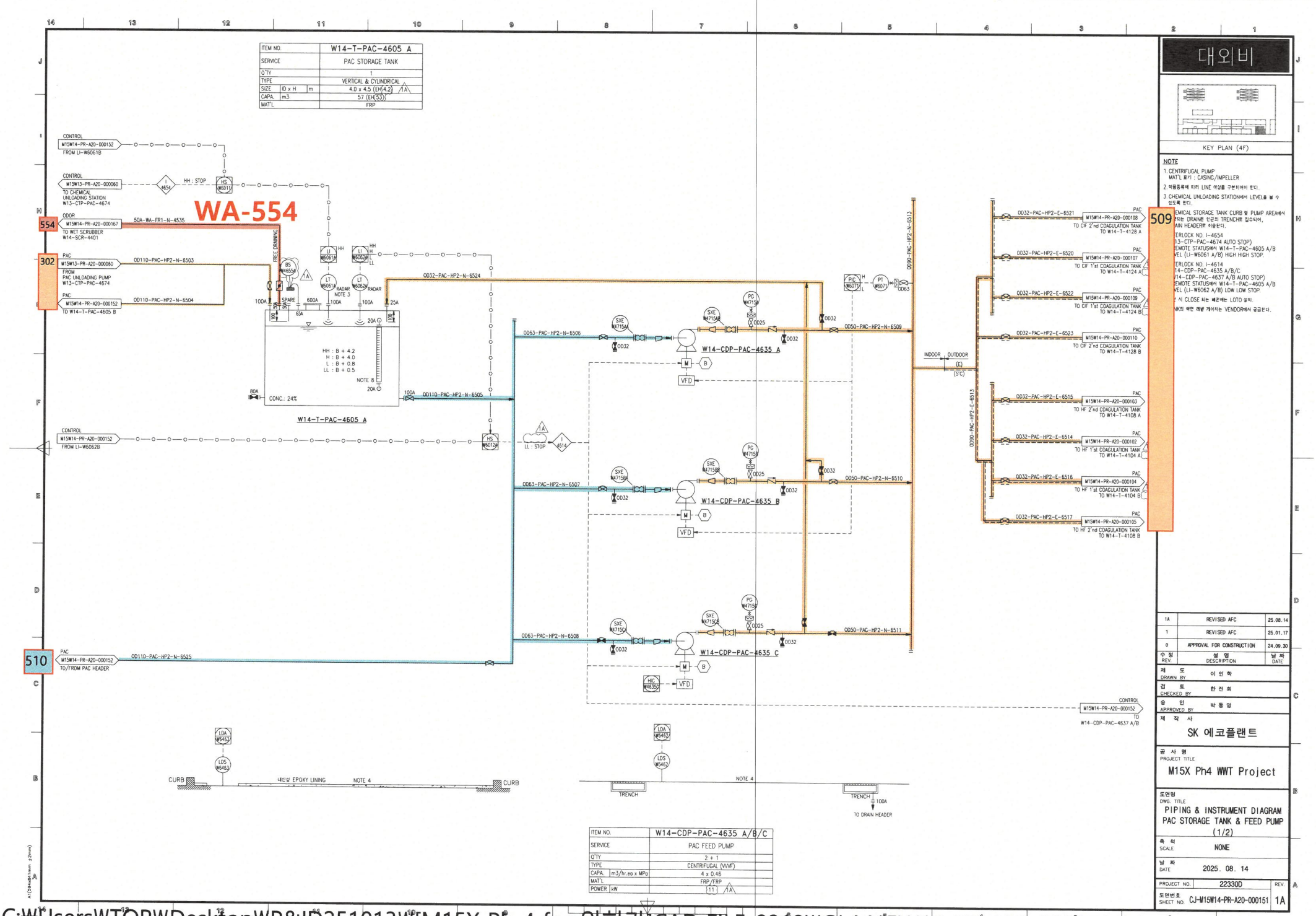
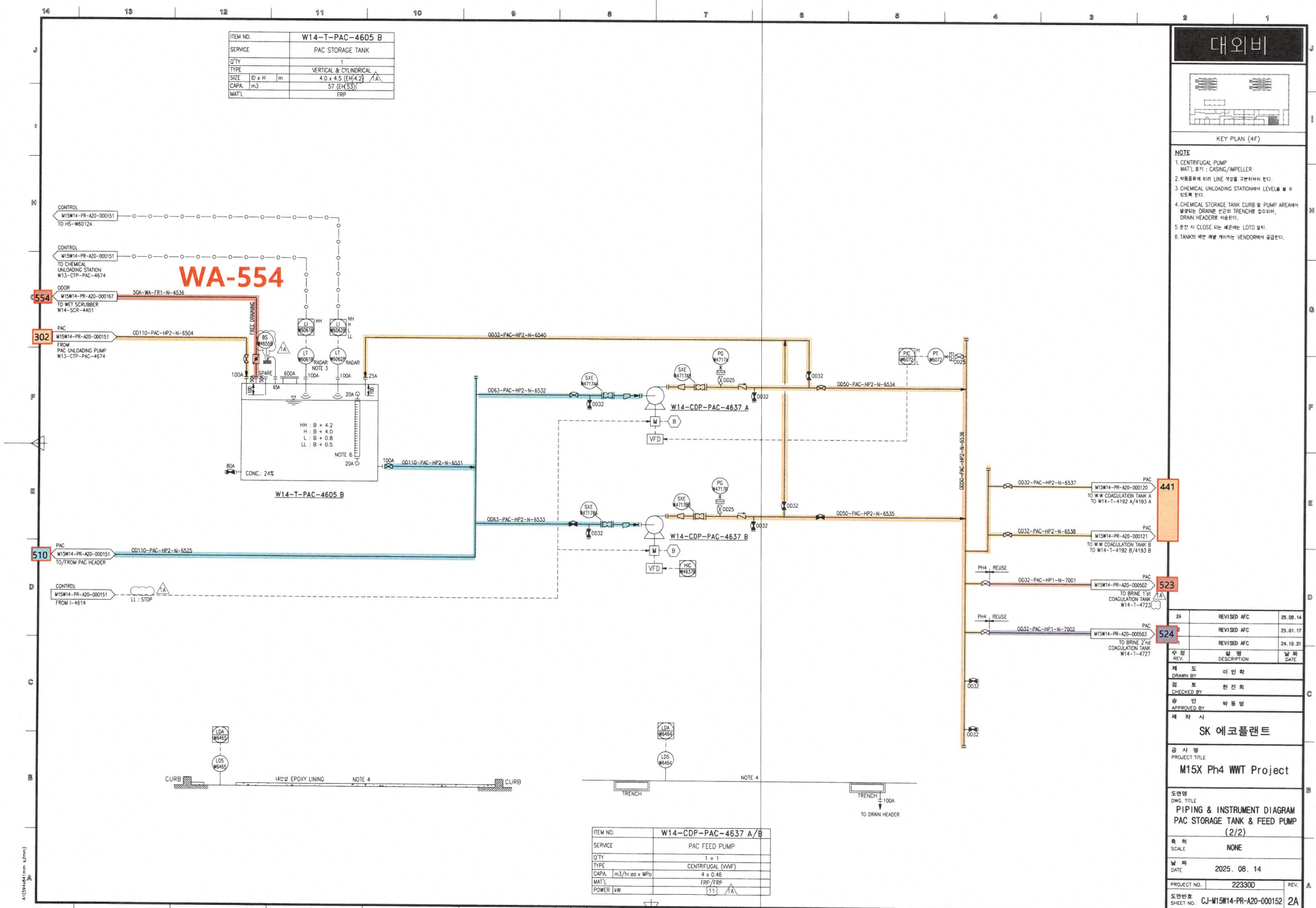


C:WUsersWTOPWDesktopWP&ID251013WM15X Ph 4 for 외허가ICAD FILE 0910WCJ-M15W14-PR-A20-000151 R1 P&ID FOR PAC S



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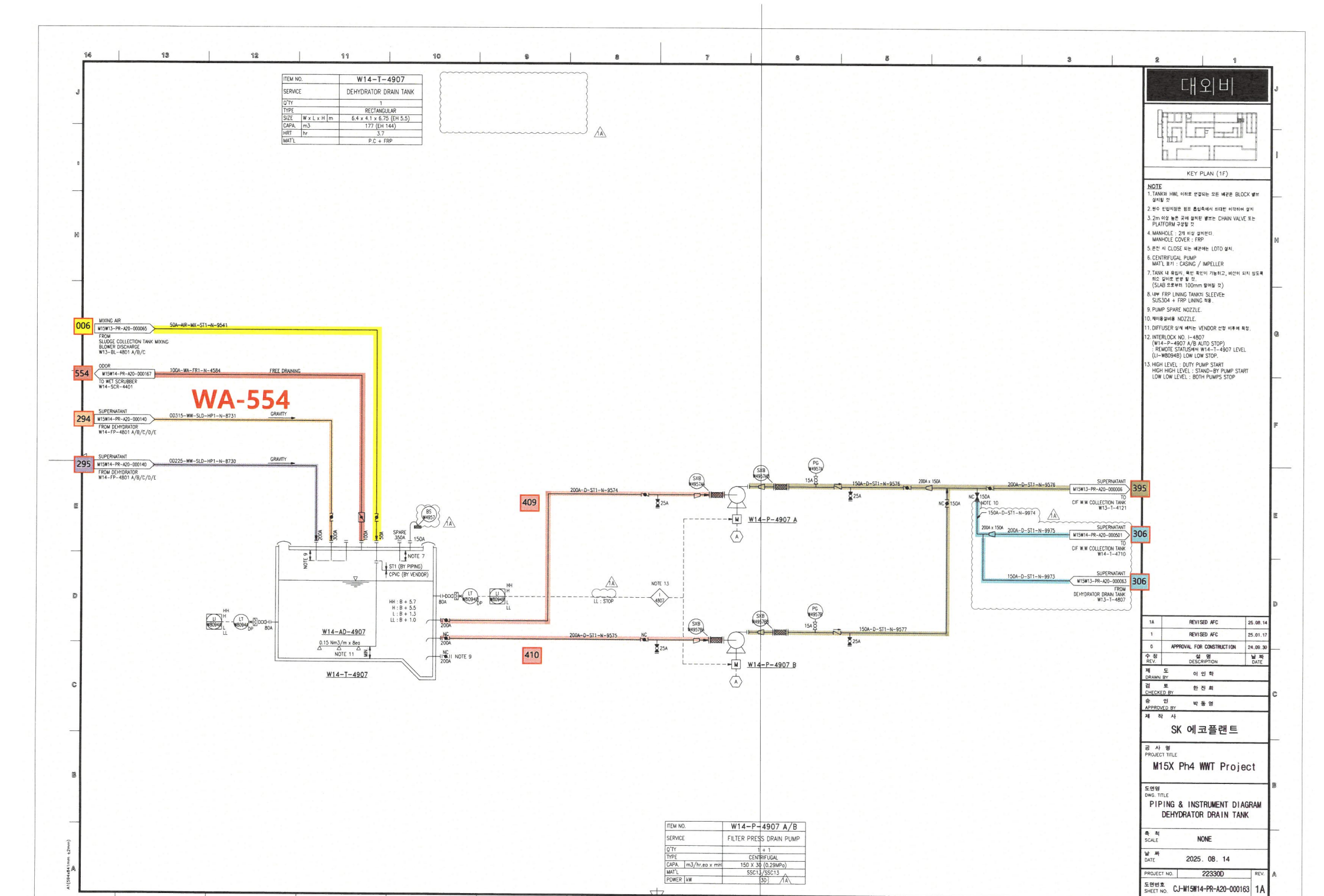

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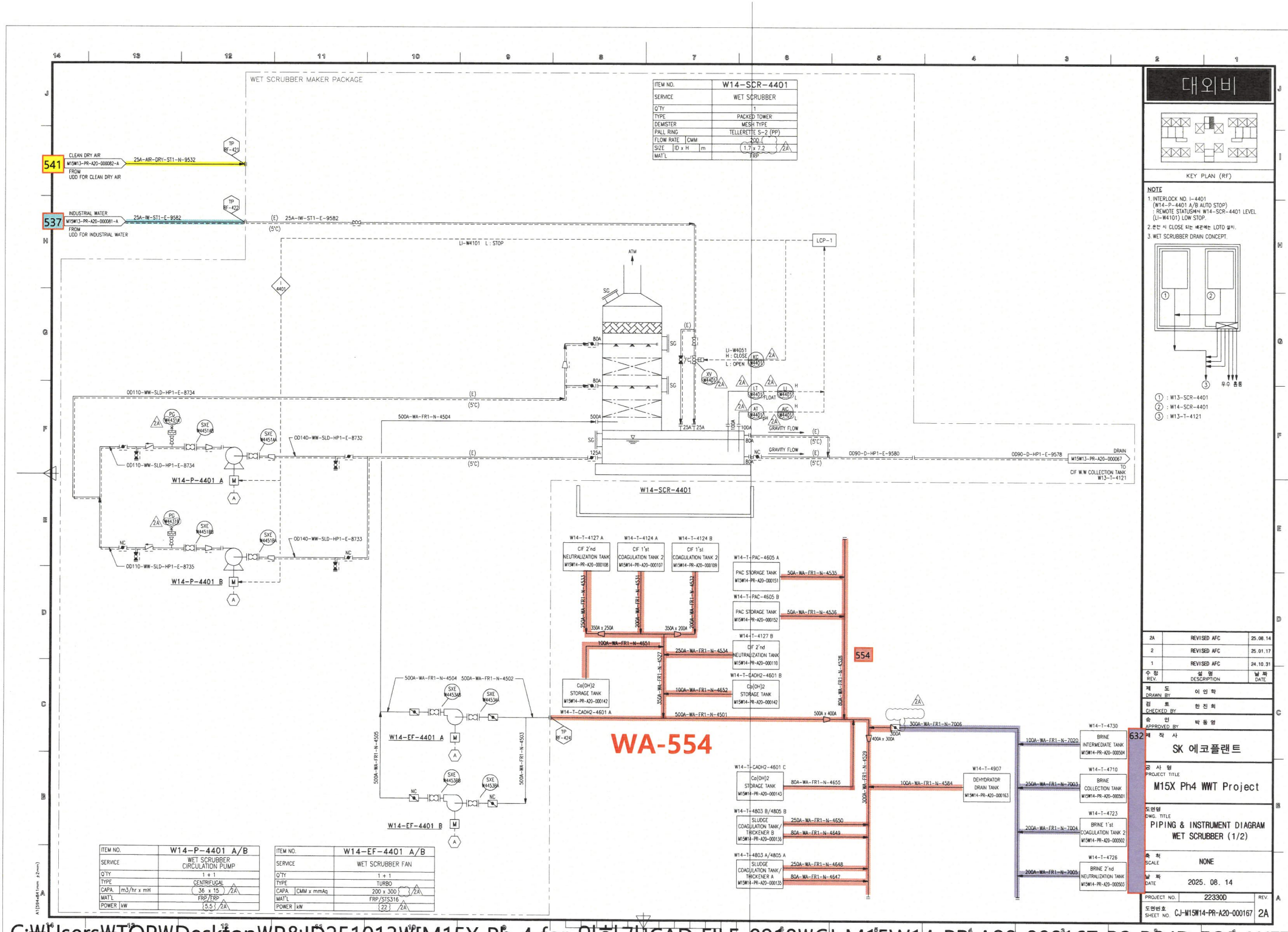


ITEM NO.	W14-P-4401 A/B
SERVICE	WET SCRUBBER CIRCULATION PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL
CAPA. m3/hr x mH	36 x 15 (2A)
MAT'L	FRP/FRP
POWER kW	(5.5) /2A

ITEM NO.	W14-EF-4401 A/B
SERVICE	WET SCRUBBER FAN
Q'TY	1 + 1
TYPE	TURBO
CAPA. CMM x mmAq	200 x 300 (2A)
MAT'L	FRP/STS316
POWER kW	(22) /2A

대외비

KEY PLAN (RF)

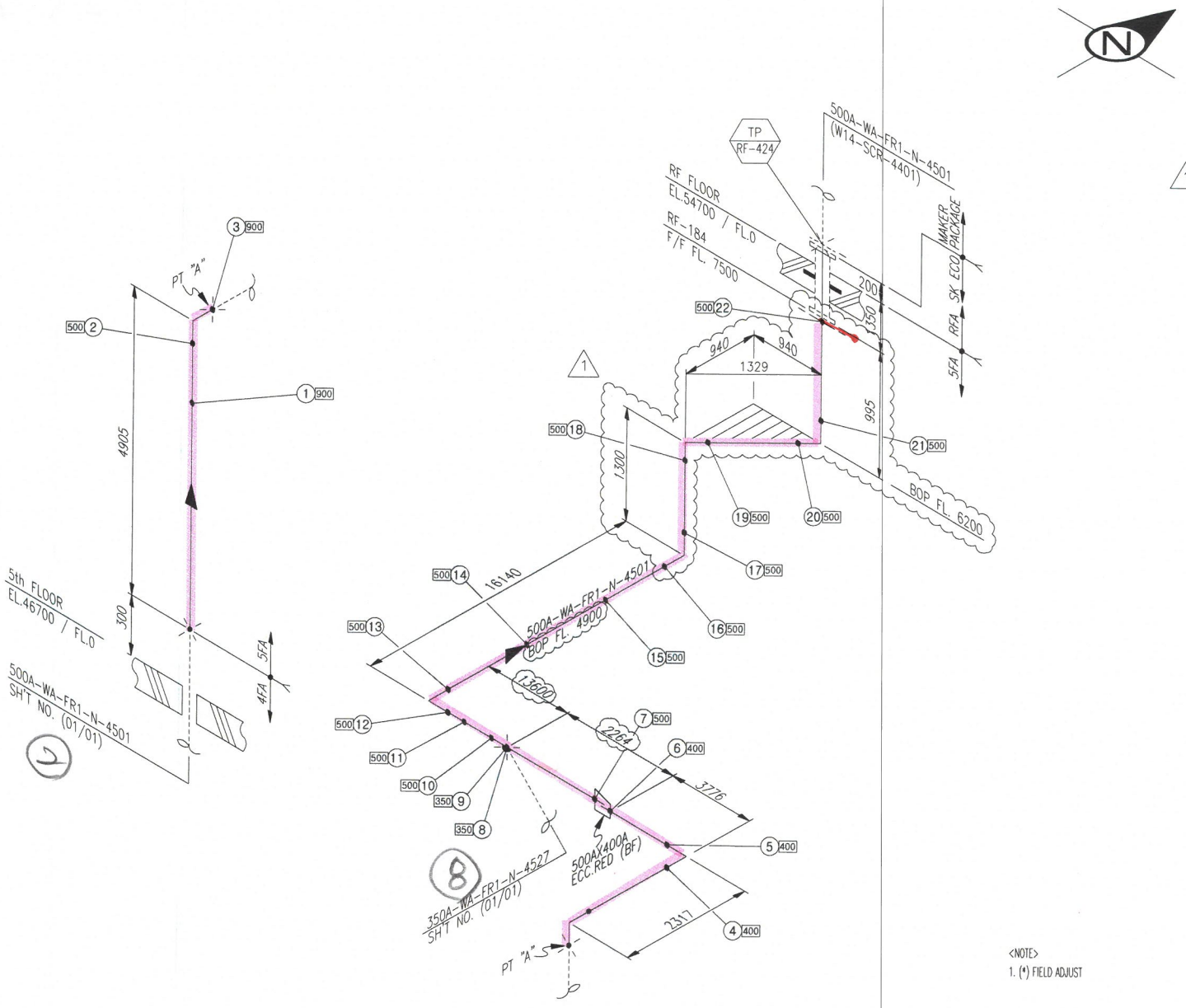
NOTE
1. INTERLOCK NO. I-4401
(W14-P-4401 A/B AUTO STOP)
: REMOTE STATUS에서 W14-SCR-4401 LEVEL
(LI-W4101) LOW STOP.
2. 운전 시 CLOSE 되는 배관에는 LDTO 설치.
3. WET SCRUBBER DRAIN CONCEPT.

① : W13-SCR-4401
② : W14-SCR-4401
③ : W13-T-4121

REV.	DESCRIPTION	DATE
2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31

DRAWN BY	이 인 학
CHECKED BY	한 진 희
APPROVED BY	박 동 영
제 작 사	SK 에코플랜트
PROJECT TITLE	M15X Ph4 WWT Project
DWG. TITLE	PIPING & INSTRUMENT DIAGRAM WET SCRUBBER (1/2)
SCALE	NONE
DATE	2025. 08. 14
PROJECT NO.	22330D
SHEET NO.	CJ-M15W14-PR-A20-000167
REV.	2A

C:₩Users₩TOPW₩Desktop₩P&ID251013₩[M15X-Ph-4 for 외허가]CAD FILE_0910₩CJ-M15W14-PR-A20-000167 R2 P&ID FOR WET S
본 도면은 SK에코플랜트(주)의 자산으로 무단 복제 및 유출을 금지합니다. 위반 시 관련 법률에 따라 민·형사상 책임을 질 수 있습니다.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	400A	10.0M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 6.4T
P	500A	31.7M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 6.4T
E90	400A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 6.4T
E90	500A	4EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 6.4T
RE	500Ax400A	1EA	REDUCER ECC, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
STU	500Ax350A	1EA	STUB-IN BRANCH
F	500A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	500A	1EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	500A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_EPDM LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1.0	AMB
			OPERATING	ATM	AMB

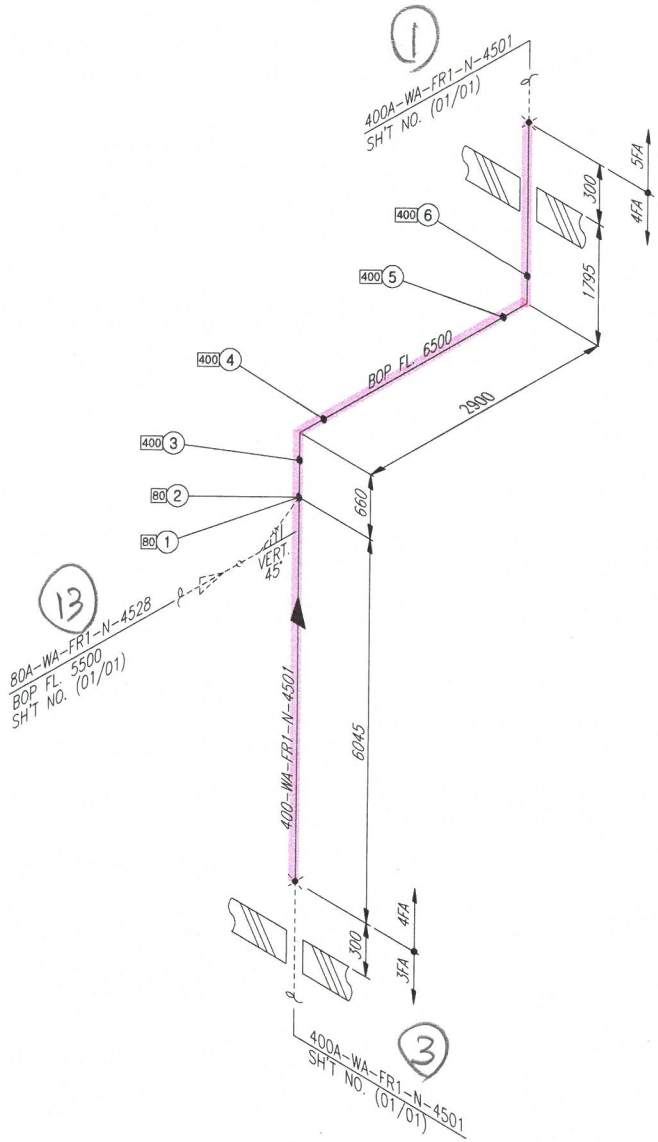
AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050005-A
REF. P & ID CJ-M15W14-PR-A20-000167

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
1	25.03.31	REVISED AS MARKED	황래종	안남후	박동영
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant
COMMODITY .WA
LINE NO. 500A-WA-FR1-N-4501
PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 1

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	400A	10.3M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
STU	400Ax80A	1EA	STUB-IN BRANCH
E90	400A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBL0/VB1/VB1L0/VB2/VB2L0 * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDL0 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1.0	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-D
REF. P & ID CJ-M15W14-PR-A20-000151

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY WA

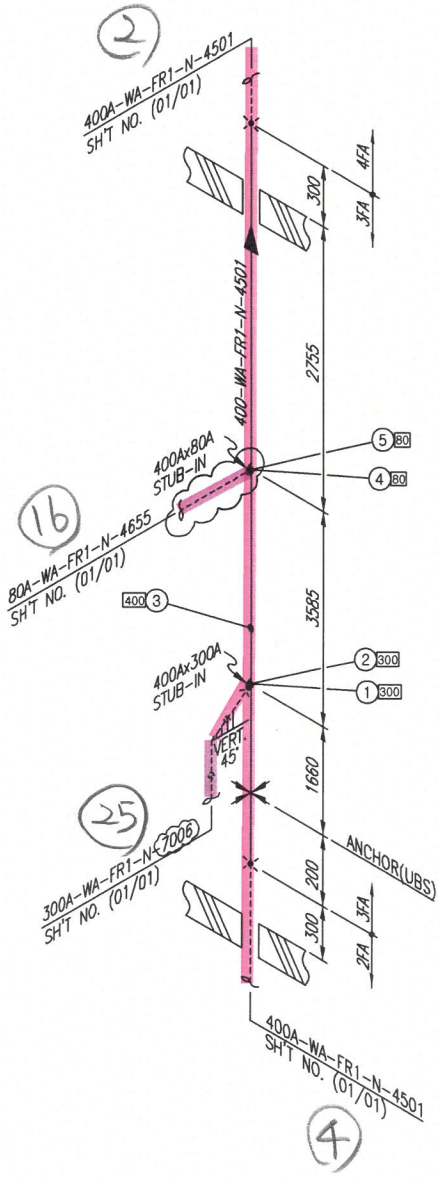
JOB NO. 22330D

LINE NO. 400A-WA-FR1-N-4501

SH'T NO. 01 / 01

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생한 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	400A	11.0M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
STU	400Ax300A	1 EA	STUB-IN BRANCH
STU	400Ax80A	1 EA	STUB-IN BRANCH

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1.0/VB1/VB11.0/VB2/VB2.0 * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VD1.0 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (3)

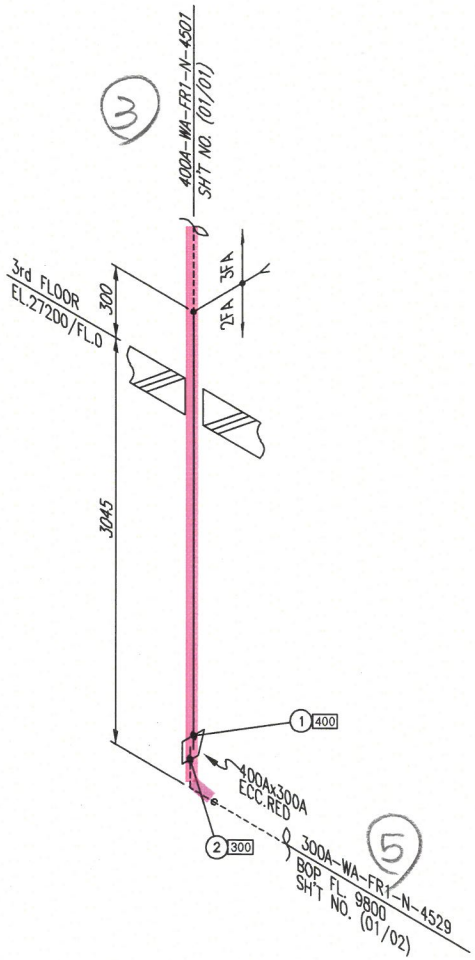
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1.0	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT 3FA
REF. PLAN DWG. CJ-M15W14-PI-A30-030007-A
REF. P & ID CJ-M15W14-PR-A20-000167

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 400A-WA-FR1-N-4501

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	400A	3.3M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
RE	400A×300A	1EA	REDUCER ECC, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

4

INSULATION	CODE	N
	THICK.	-

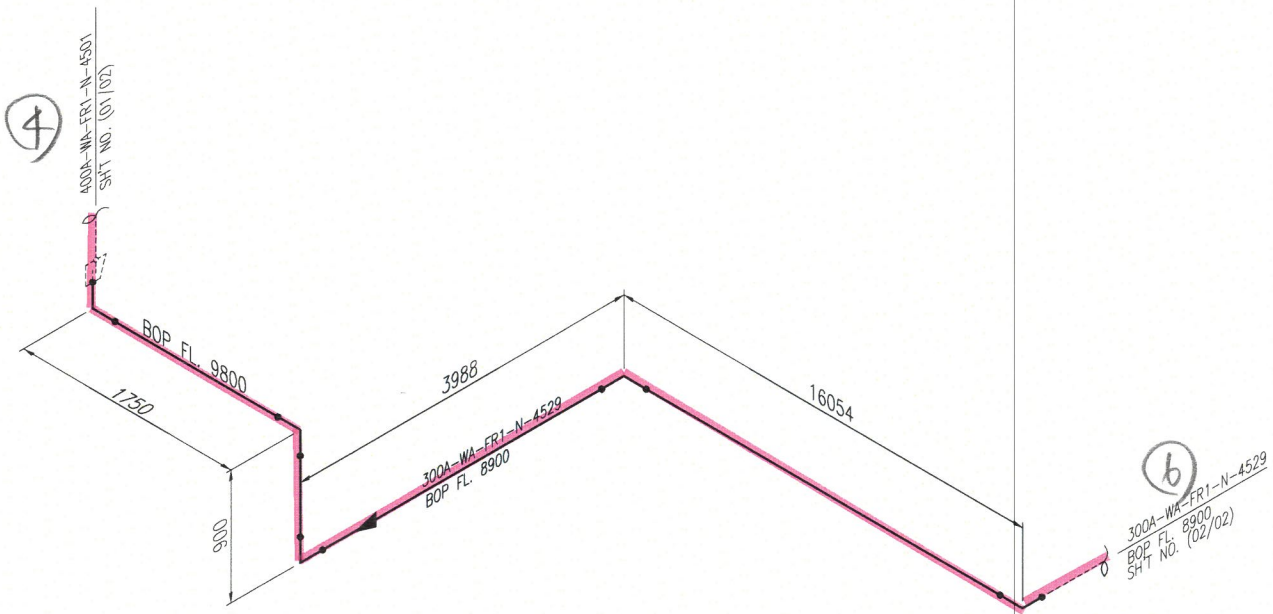
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	1	AMB
OPERATING	ATM	AMB

AREA W14
UNIT 2FA
REF. PLAN DWG. CJ-M15W14-PI-A30-020012-H
REF. P & ID CJ-M15W14-PR-A20-000167

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	24.12.20	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 400A-WA-FR1-N-4501

PROJECT NAME 400A-WA-FR1-N-4501
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	300A	22.7M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	300A	5EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	1	AMB
OPERATING	ATM	AMB

AREA W14
UNIT 2FA
REF. PLAN DWG. CJ-M15W14-PI-A30-020007-G
REF. P & ID CJ-M15W14-PR-A20-000167

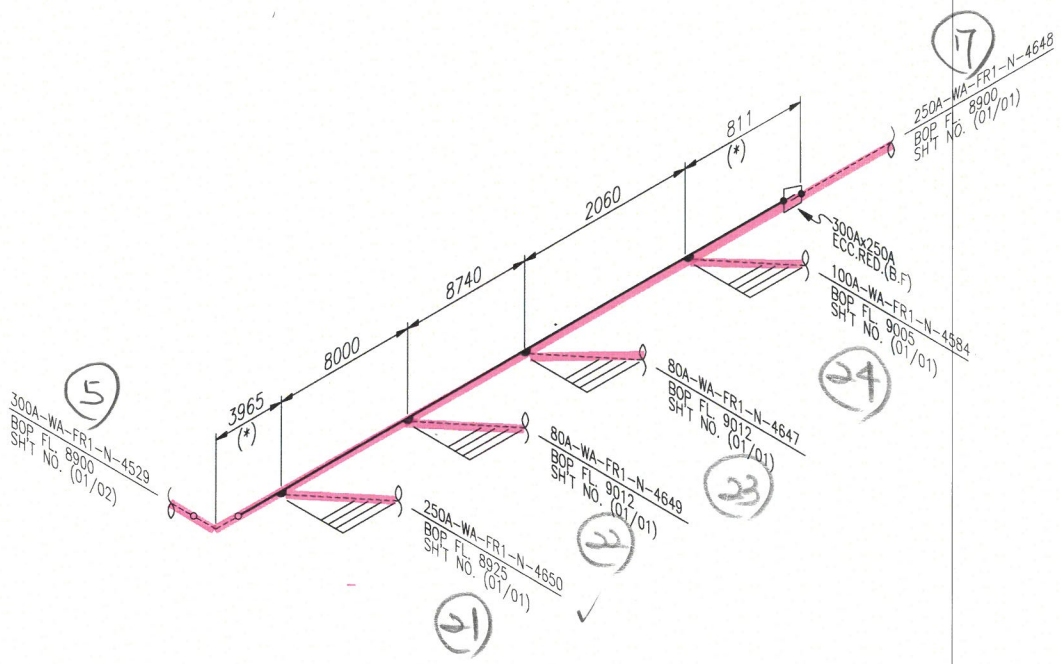
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 300A-WA-FR1-N-4529

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 02
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	900A	23.6M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
RE	300Ax250A	1EA	REDUCER CON, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
STU	300Ax250A	1EA	STUB-IN BRANCH
STU	300Ax100A	1EA	STUB-IN BRANCH
STU	300Ax80A	2EA	STUB-IN BRANCH

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	1	AMB
OPERATING	ATM	AMB

AREA	W14
UNIT	2FA
REF. PLAN DWG.	CJ-M15W14-PI-A30-020008-G
REF. P & ID	CJ-M15W14-PR-A20-000167

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	24.12.20	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT: SK hynix

CONTRACTOR: SK ecoplant

PROJECT NAME: M15X Ph4 Project

COMMODITY: WA

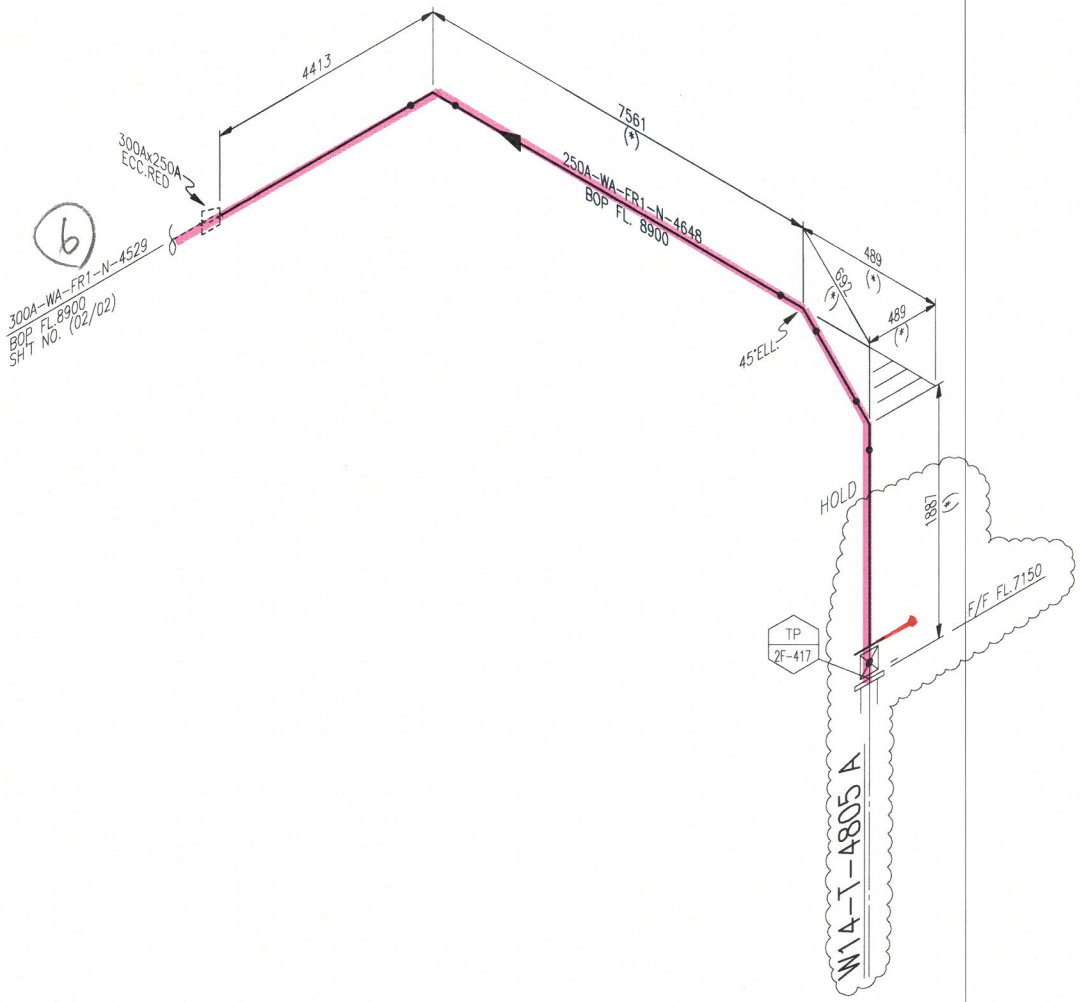
JOB NO.: 22330D

LINE NO.: 300A-WA-FR1-N-4529

SH'T NO.: 02 / 02

REV.: 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	250A	14.6M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	250A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
E45	250A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
VP	250A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
F	250A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	250A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	250A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBL0/VB1/VB1L0/VB2/VB2L0 * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDL0 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL EPDM LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (7)

INSULATION	CODE	N
	THICK.	–

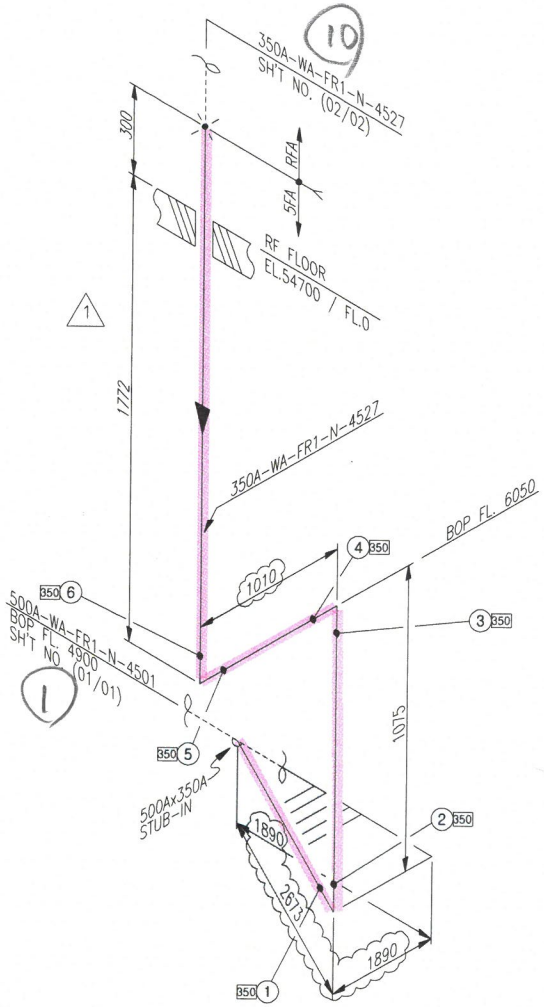
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	1	AMB
OPERATING	ATM	AMB

AREA W14
UNIT 2FA
REF. PLAN DWG. CJ-M15W14-PI-A30-020003-G
REF. P & ID CJ-M15W14-PR-A20-000135

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	24.12.20	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY WA
JOB NO. 22330D
LINE NO. 250A-WA-FR1-N-4648
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	350A	4.9M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	350A	3EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050009-C
REF. P & ID CJ-M15W14-PR-A20-000167

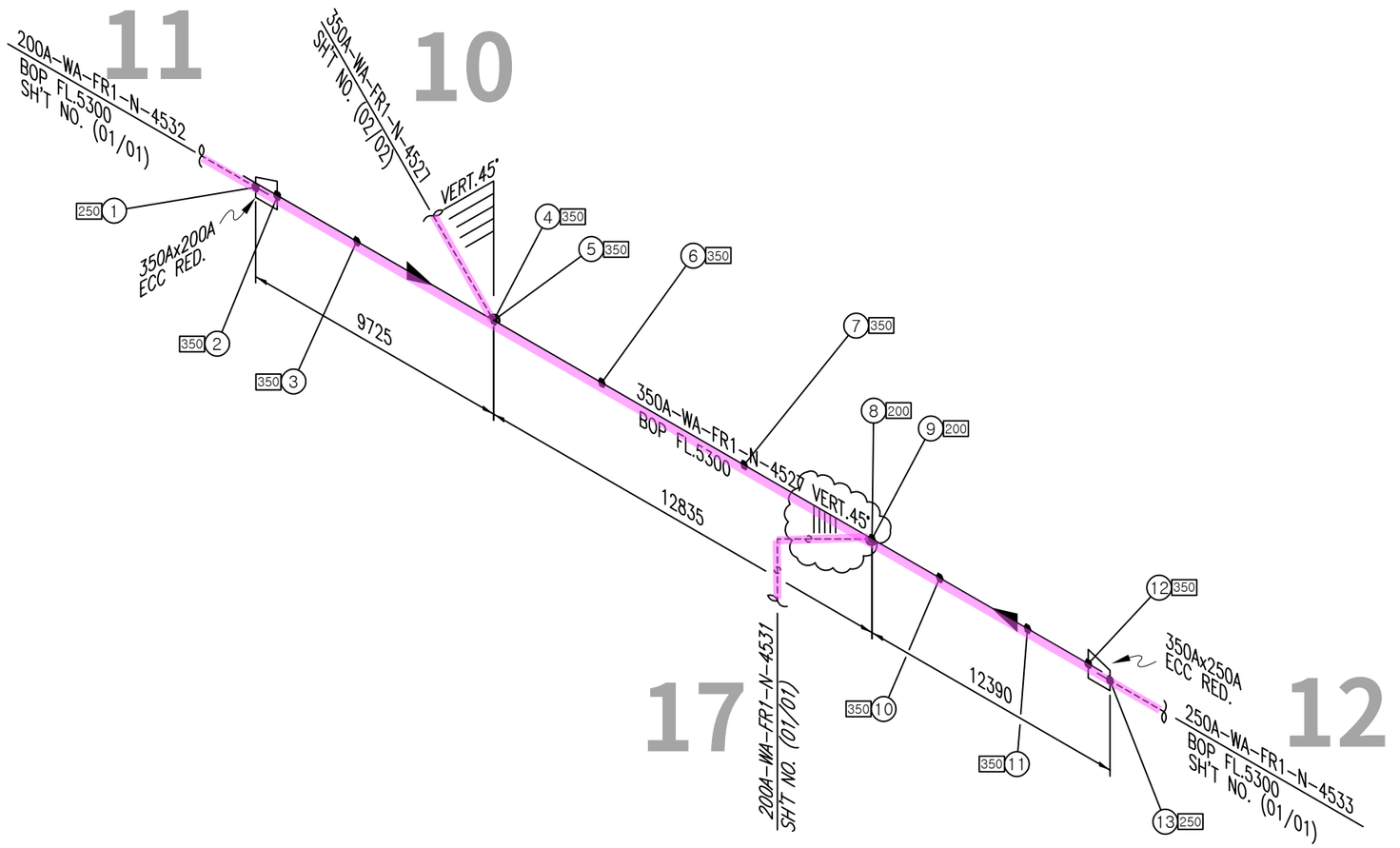
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
1	25.03.31	REVISED AS MARKED	황래종	안남후	박동영
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 350A-WA-FR1-N-4527

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 1

CLIENT SK hynix

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
	FR1		
P	350A	34.3M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
RE	350Ax250A	1EA	REDUCER ECC, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
RE	350Ax200A	1EA	REDUCER ECC, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
STU	350Ax200A	1EA	STUB-IN BRANCH
STU	350Ax350A	1EA	STUB-IN BRANCH

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	1	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10001-B
REF. P & ID CJ-M15W14-PR-A20-000167

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

9

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY WA

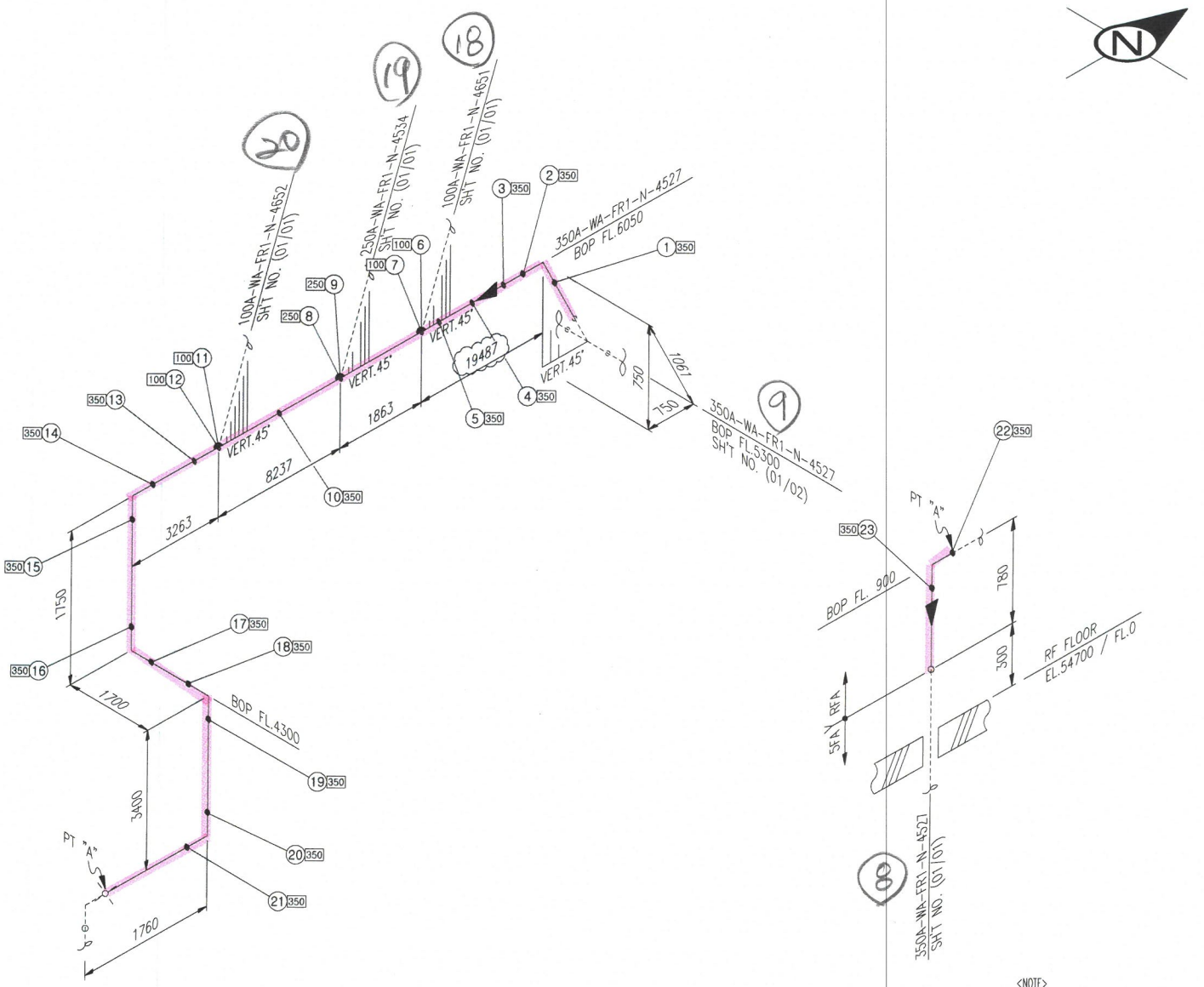
JOB NO. 22330D

LINE NO. 350A-WA-FR1-N-4527

SH'T NO. 01 / 02

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	350A	52.0M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	350A	4EA	ELBOW 90 DEG SR, STS304, ERW, BE - SCH 10S
E45	350A	1EA	ELBOW 45 DEG LR, STS304, ERW, BE - SCH 10S
STU	350Ax250A	1EA	STUB-IN BRANCH
STU	350Ax100A	2EA	STUB-IN BRANCH

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10001-B
REF. P & ID CJ-M15W14-PR-A20-000167

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY WA

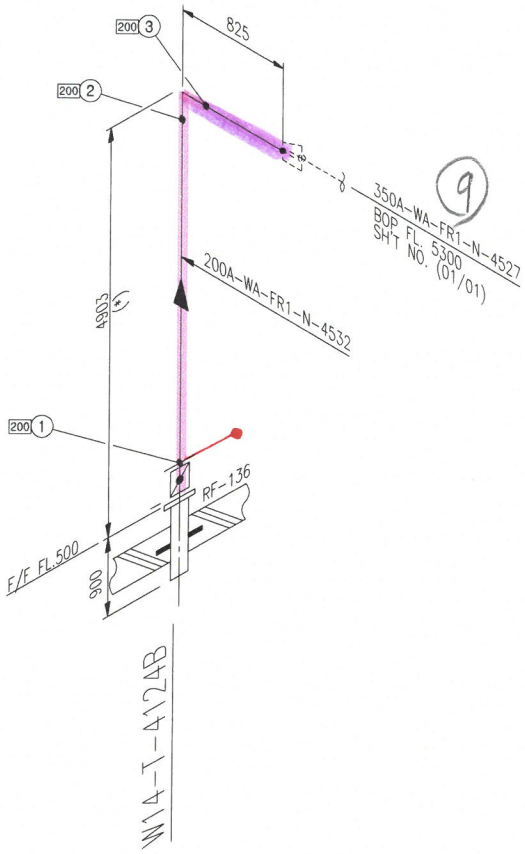
JOB NO. 22330D

LINE NO. 350A-WA-FR1-N-4527

SH'T NO. 02 / 02

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	200A	4.9M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	200A	1EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	200A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	200A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	200A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VP	200A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (11)

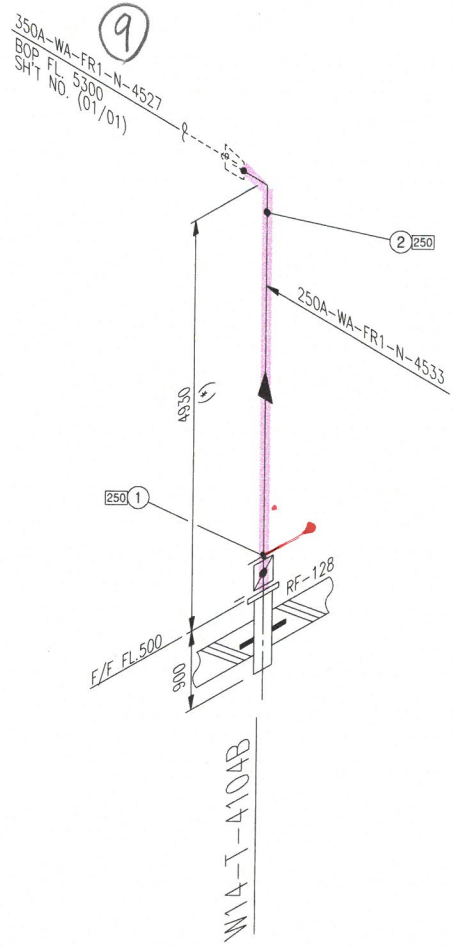
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10003-A
REF. P & ID CJ-M15W14-PR-A20-000109

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 200A-WA-FR1-N-4532
PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	250A	5.0M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	250A	1EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	250A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	250A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	250A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VP	250A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (12)

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1	AMB
			OPERATING	ATM	AMB

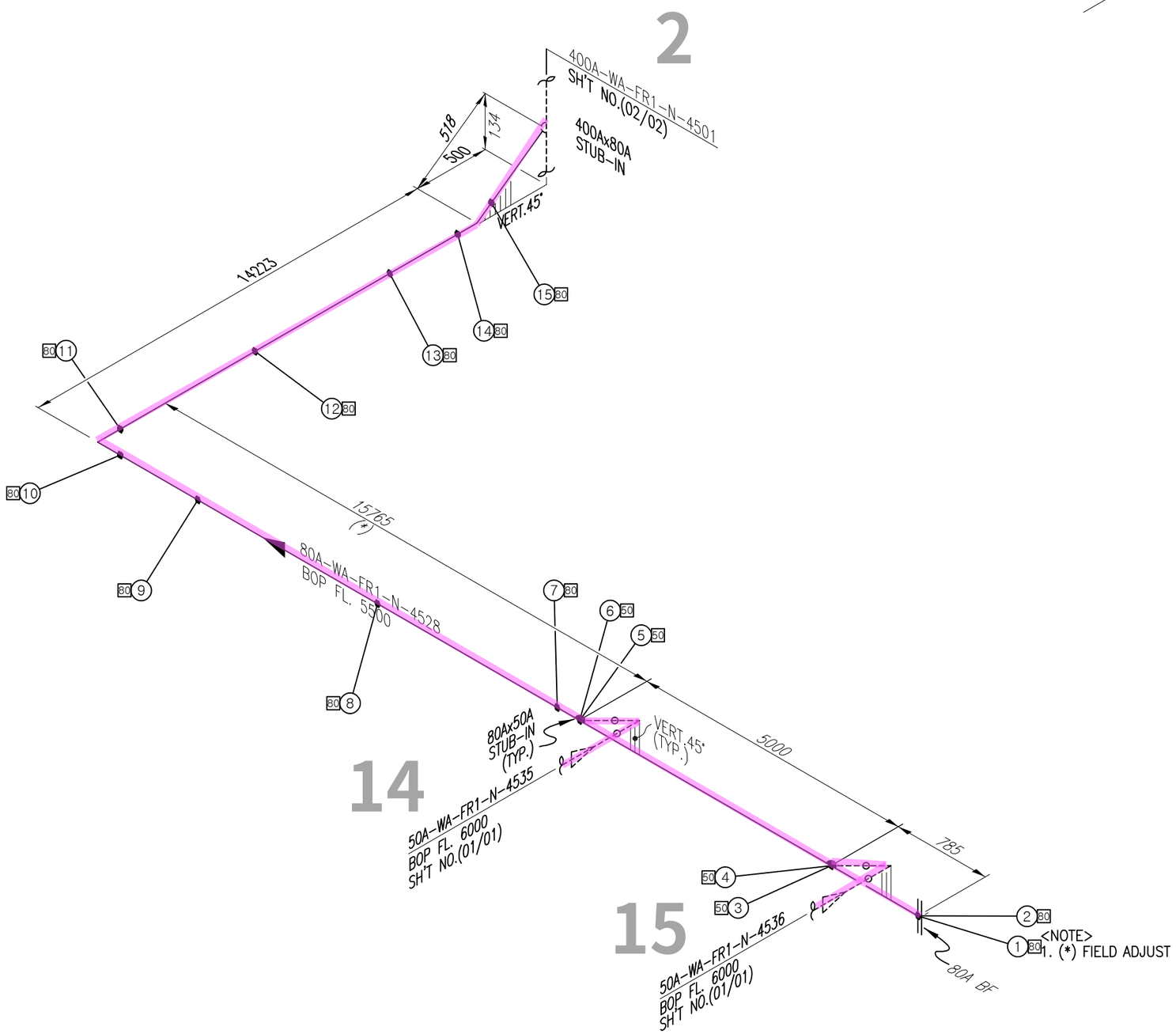
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10001-A
REF. P & ID CJ-M15W14-PR-A20-000108

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 250A-WA-FR1-N-4533

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	80A	36.4M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
STU	80Ax50A	2EA	STUB-IN BRANCH
F	80A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	80A	1EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	80A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
E90	80A	1EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
E45	80A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBL0/VB1/VB1L0/VB2/VB2L0 * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDL0 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1.0	AMB
			OPERATING	ATM	AMB

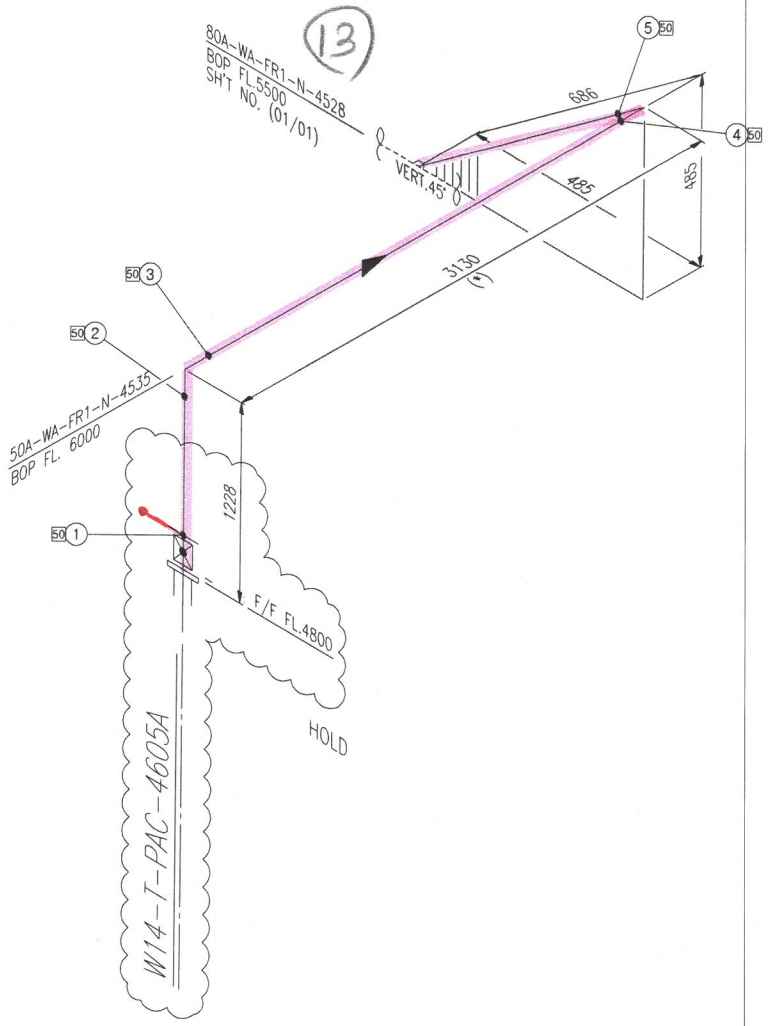
AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-D
REF. P & ID CJ-M15W14-PR-A20-000151

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 80A-WA-FR1-N-4528

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	50A	5.0M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
E90	50A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
F	50A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	50A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	50A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VP	50A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1.0	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040013-D
REF. P & ID CJ-M15W14-PR-A20-000151

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래즘	안남후	박동영

CLIENT SK hynix (14)

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY WA

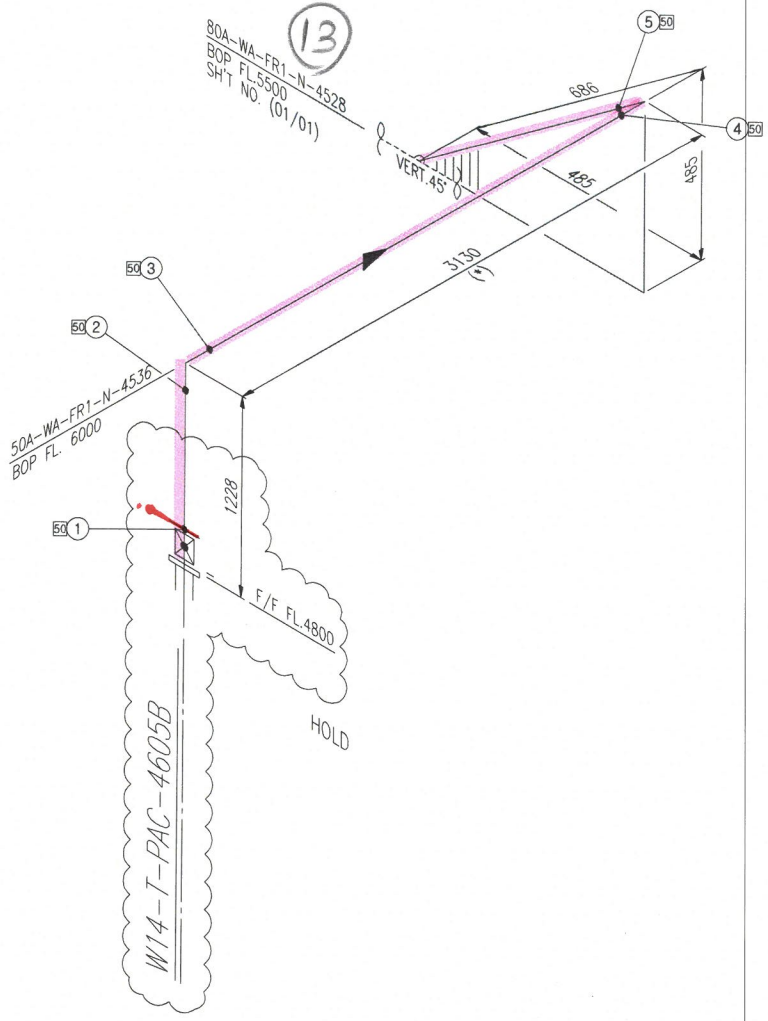
JOB NO. 22330D

LINE NO. 50A-WA-FR1-N-4535

SH'T NO. 01 / 01

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생한 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	50A	5.0M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
E90	50A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
F	50A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	50A	1EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	50A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VP	50A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1.0	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT 4FA
REF. PLAN DWG. CJ-M15W14-PI-A30-040014-D
REF. P & ID CJ-M15W14-PR-A20-000152

REV. NO.	DATE	DESCRIPTION	황래중 BY	안남후 CHK	박동영 APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION			

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 50A-WA-FR1-N-4536

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

CLIENT SK hynix

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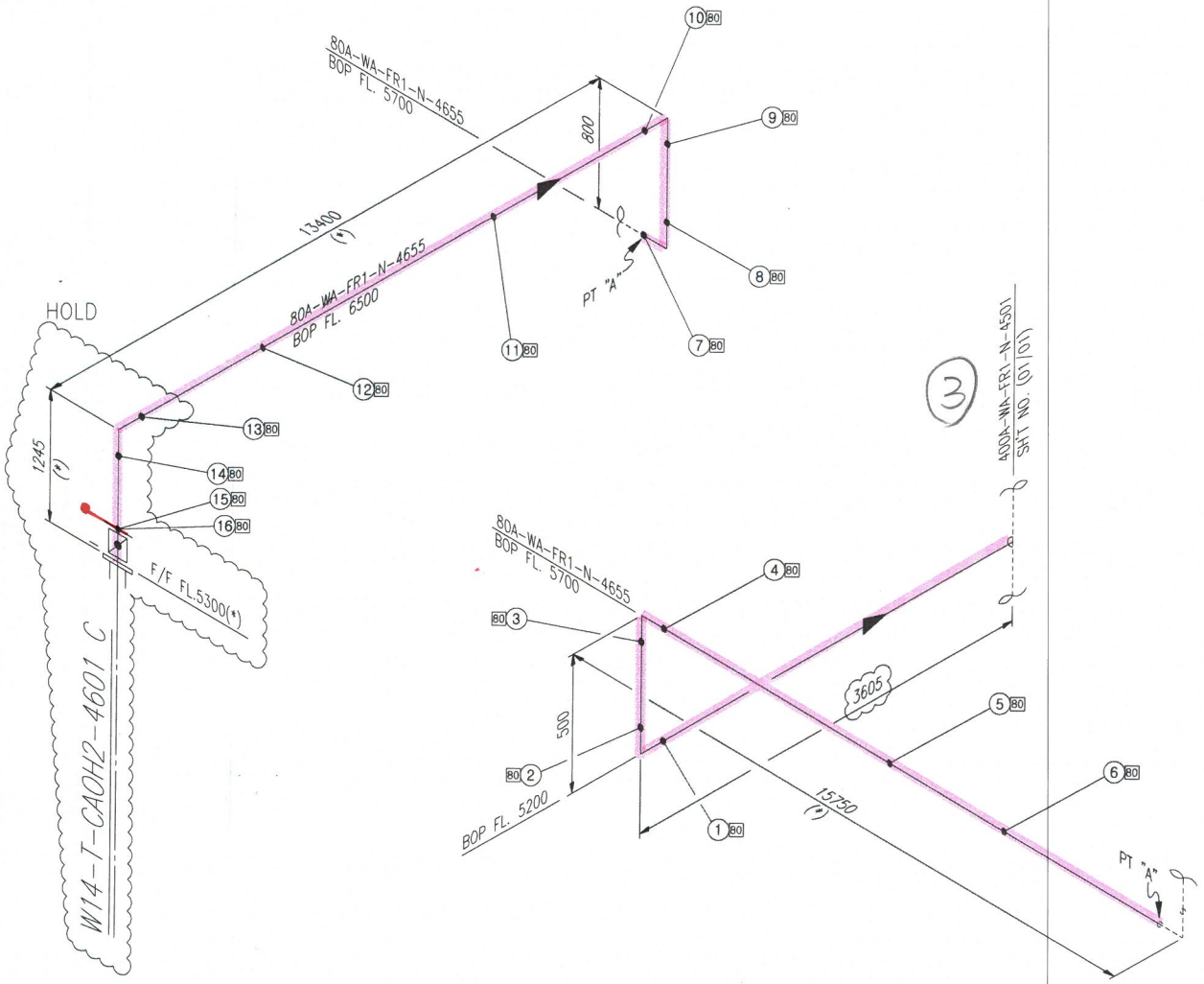
경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	80A	36.1M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
E90	80A	5EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
F	80A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	80A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	200A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VP	150A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBL0/VB1/VB1L0/VB2/VB2L0 * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDL0 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT 3FA
REF. PLAN DWG. CJ-M15W14-PI-A30-030007-D
REF. P & ID CJ-M15W14-PR-A20-000143

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY WA

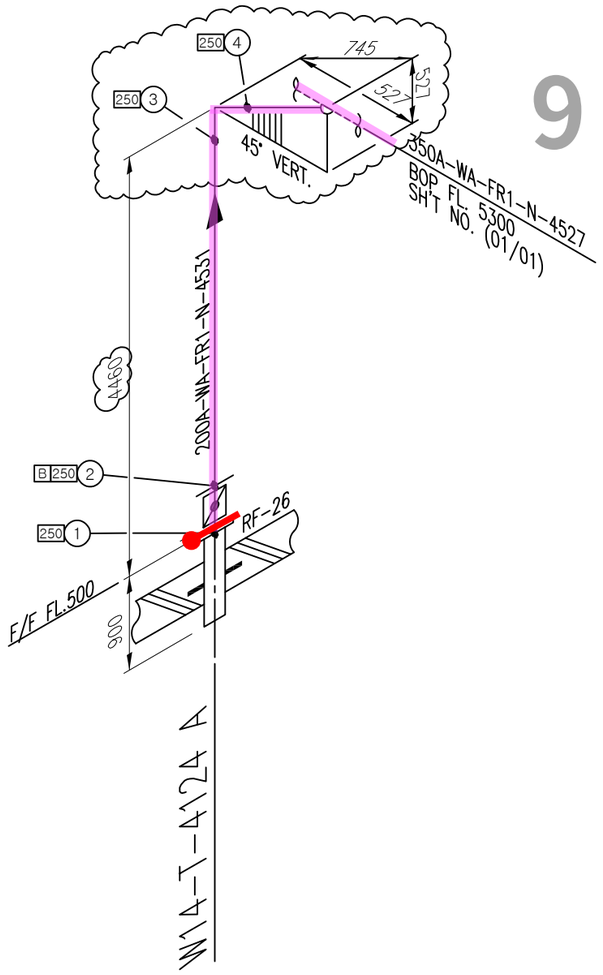
JOB NO. 22330D

LINE NO. 80A-WA-FR1-N-4655

SH'T NO. 01 / 01

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	200A	5.2M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
~~E90~~	~~200A~~	~~3EA~~	~~ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T~~
F	200A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	200A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	200A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VP	200A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
E45	200A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

17

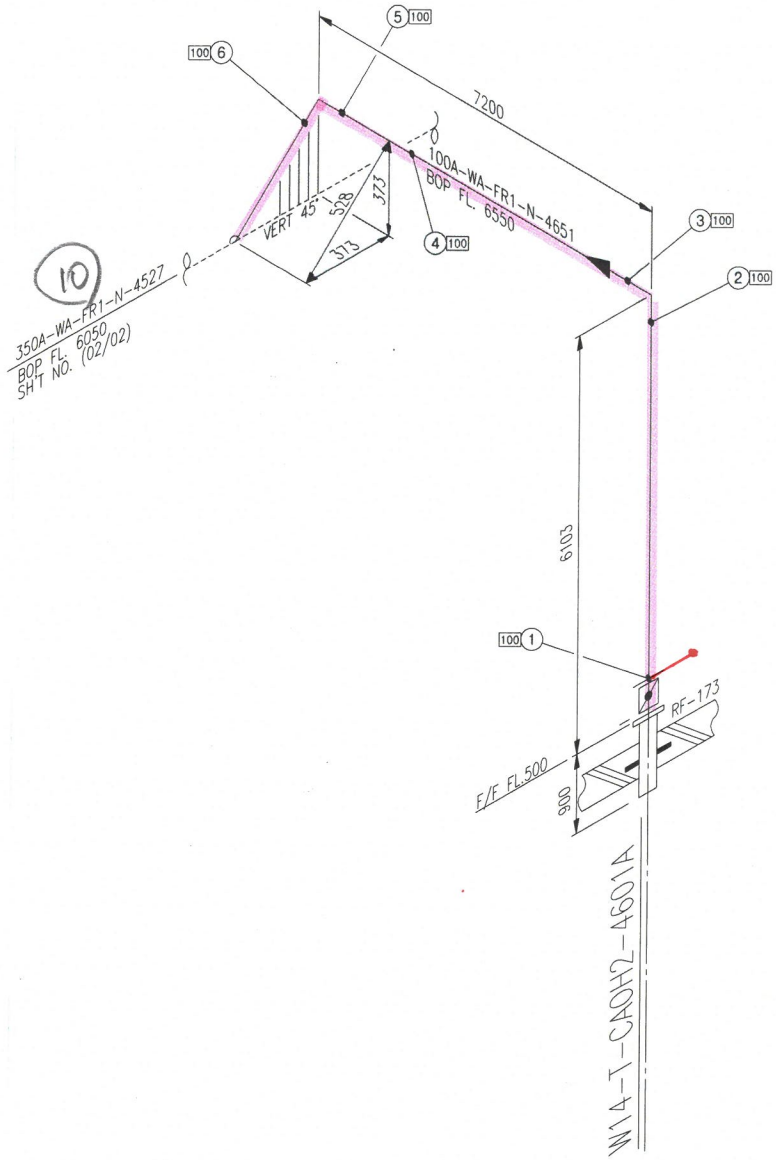
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	1	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-A
REF. P & ID CJ-M15W14-PR-A20-000167

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY WA
JOB NO. 22330D
LINE NO. 200A-WA-FR1-N-4531
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	100A	14.4M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	100A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	100A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	100A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	100A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VP	100A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

18

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1	AMB
			OPERATING	ATM	AMB

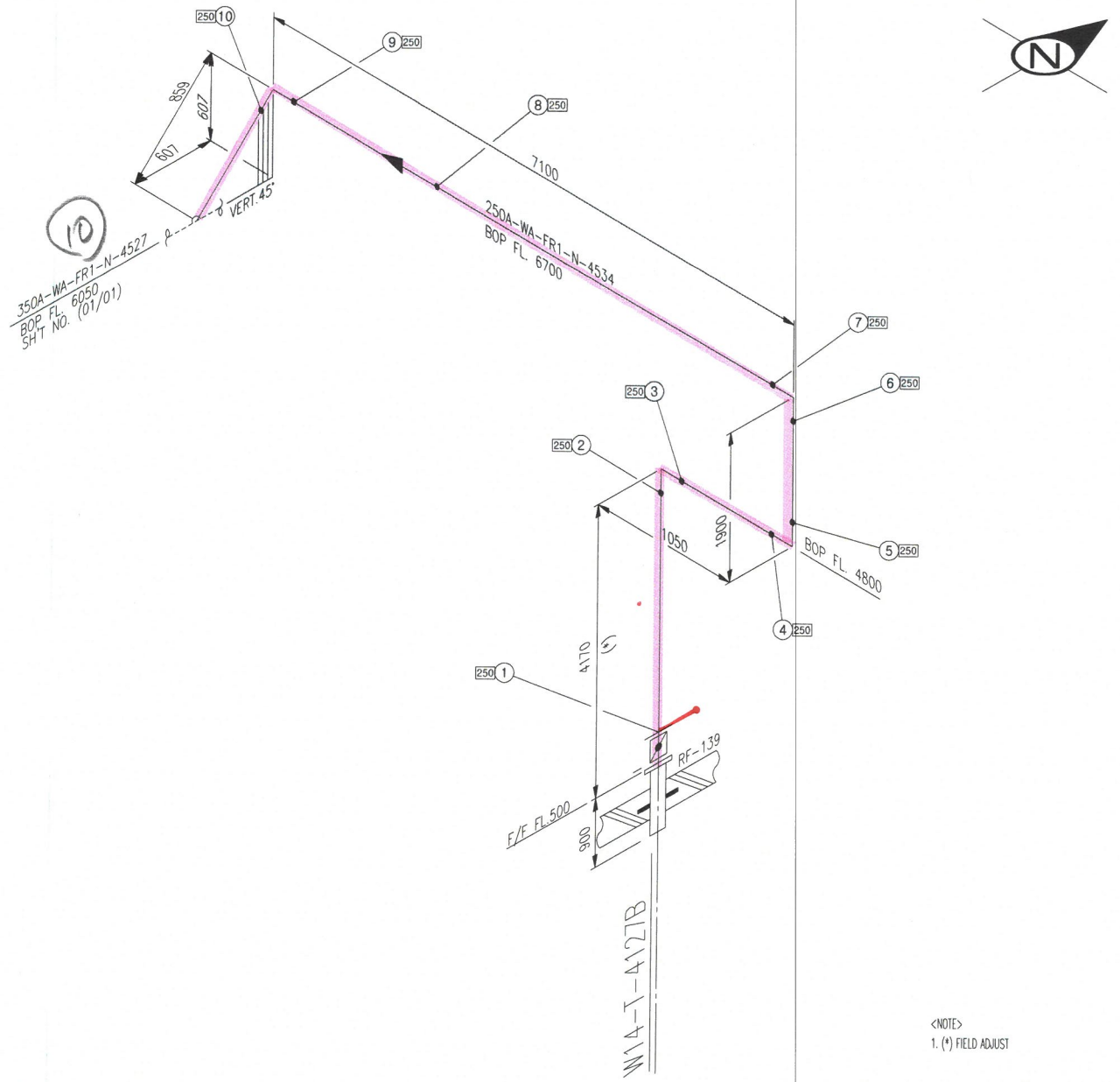
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10005-A
REF. P & ID CJ-M15W14-PR-A20-000142

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 100A-WA-FR1-N-4651

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	250A	12.4M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	250A	4EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	250A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	250A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	250A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VP	250A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10005-A
REF. P & ID CJ-M15W14-PR-A20-000110

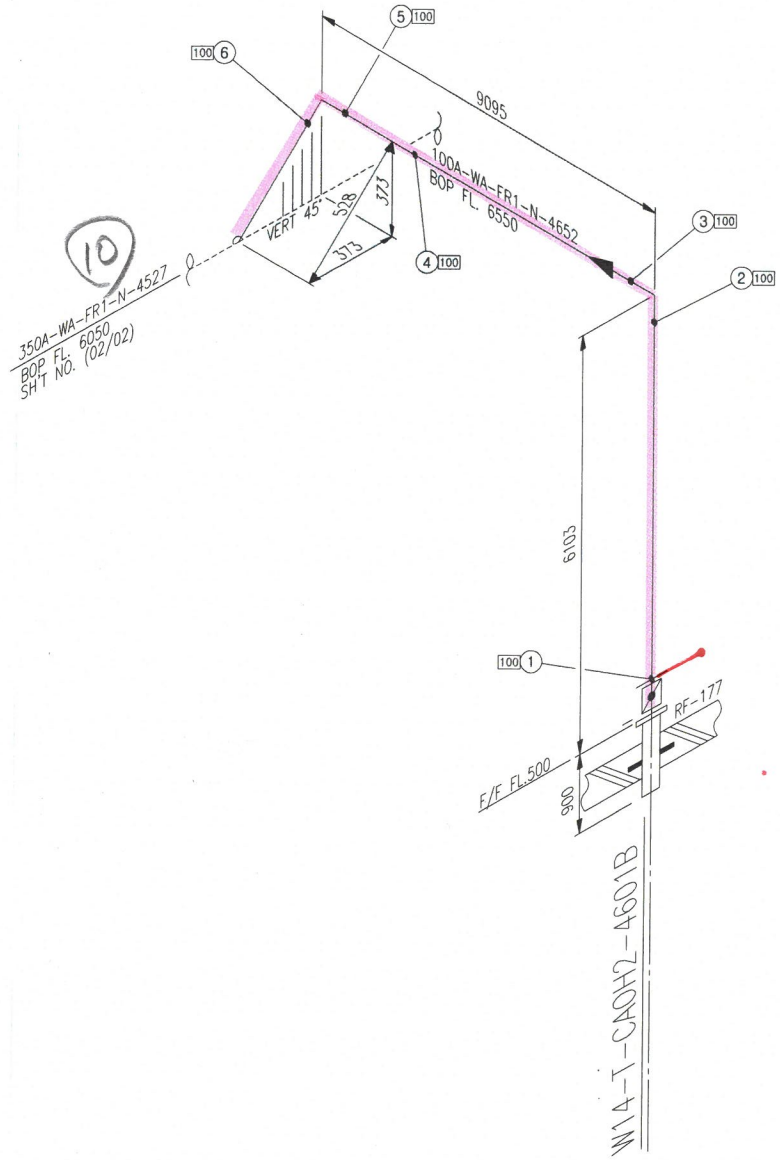
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 250A-WA-FR1-N-4534

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



N

<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	100A	14.3M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	100A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	100A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	100A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	100A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VP	100A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

20

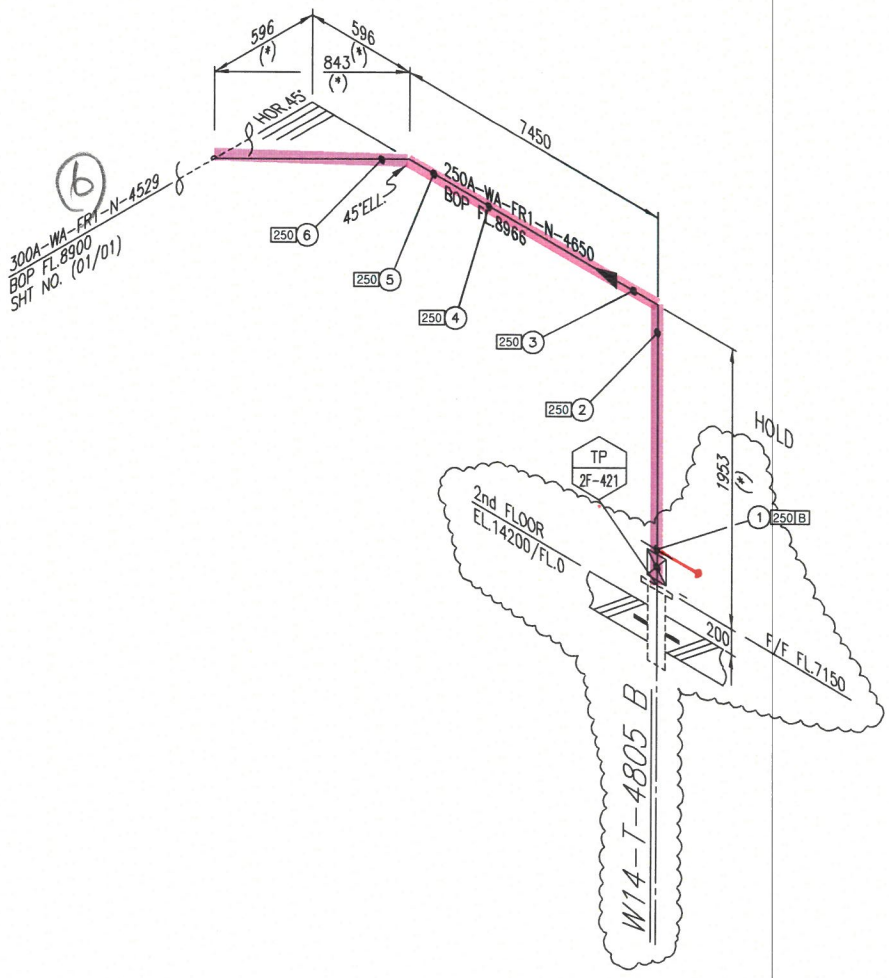
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	1	AMB
			OPERATING	ATM	AMB

AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10005-A
REF. P & ID CJ-M15W14-PR-A20-000142

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 100A-WA-FR1-N-4652
PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	250A	10.1M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	250A	1EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
E45	250A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
VP	250A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
F	250A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	250A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM THK
B	250A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1.0/VB1/VB11.0/VB2/VB21.0 * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VD1.0 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL EPDM LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTD : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (21)

INSULATION	CODE	N
	THICK.	-

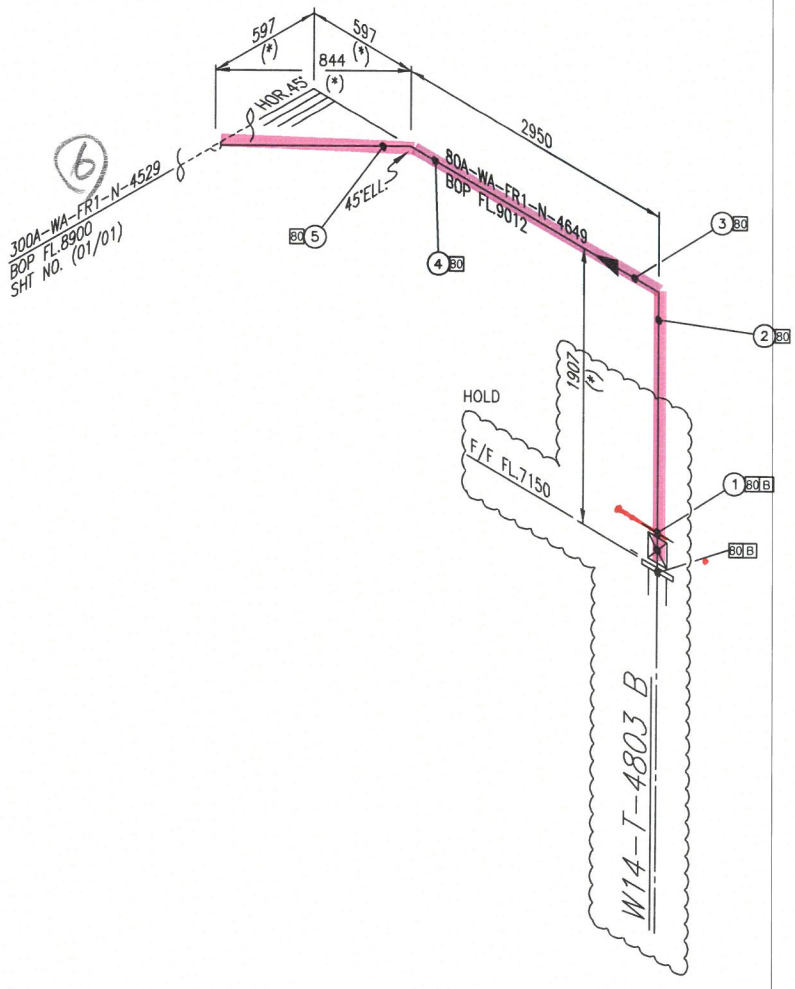
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	1	AMB
OPERATING	ATM	AMB

AREA W14
UNIT 2FA
REF. PLAN DWG. CJ-M15W14-PI-A30-020008-G
REF. P & ID CJ-M15W14-PR-A20-000136

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 250A-WA-FR1-N-4650

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	80A	5.7M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
E90	80A	1EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
E45	80A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
VP	80A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
F	80A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	80A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	80A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90 ELBOW : E90 * 45 ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL EPDM LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

INSULATION	CODE	N
	THICK.	-

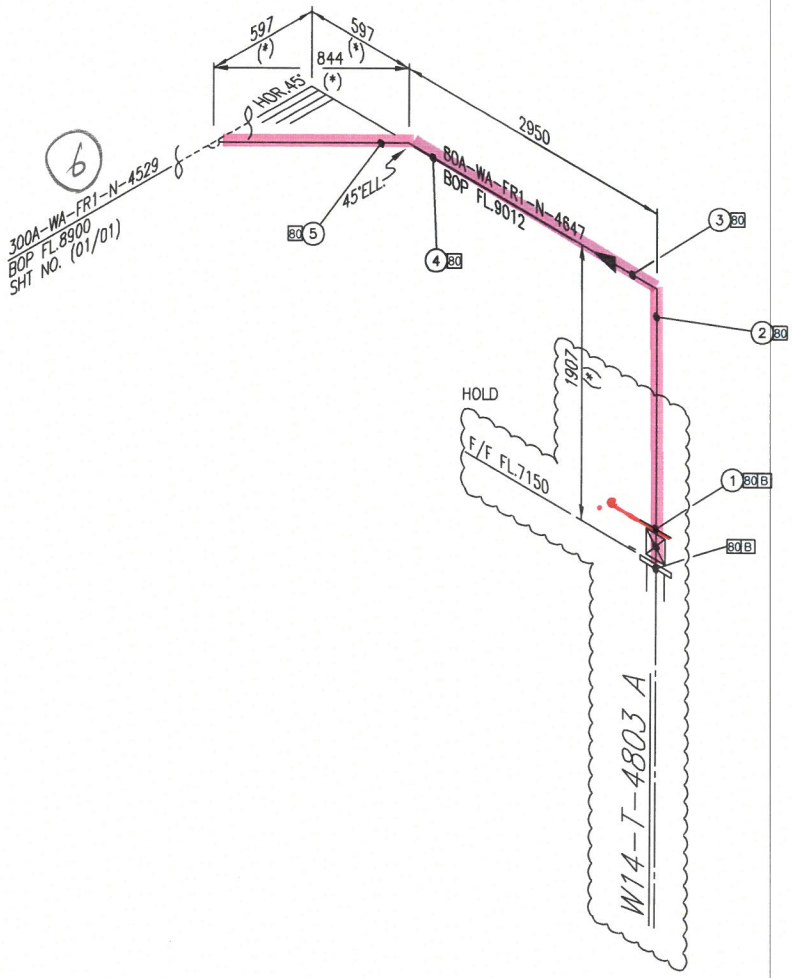
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	1	AMB
OPERATING	ATM	AMB

AREA W14
UNIT 2FA
REF. PLAN DWG. CJ-M15W14-PI-A30-020008-G
REF. P & ID CJ-M15W14-PR-A20-000136

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 80A-WA-FR1-N-4649

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	80A	5.7M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
E90	80A	1EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
E45	80A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
VP	80A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
F	80A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	80A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM THK
B	80A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL EPDM LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (23)

INSULATION	CODE	N
	THICK.	-

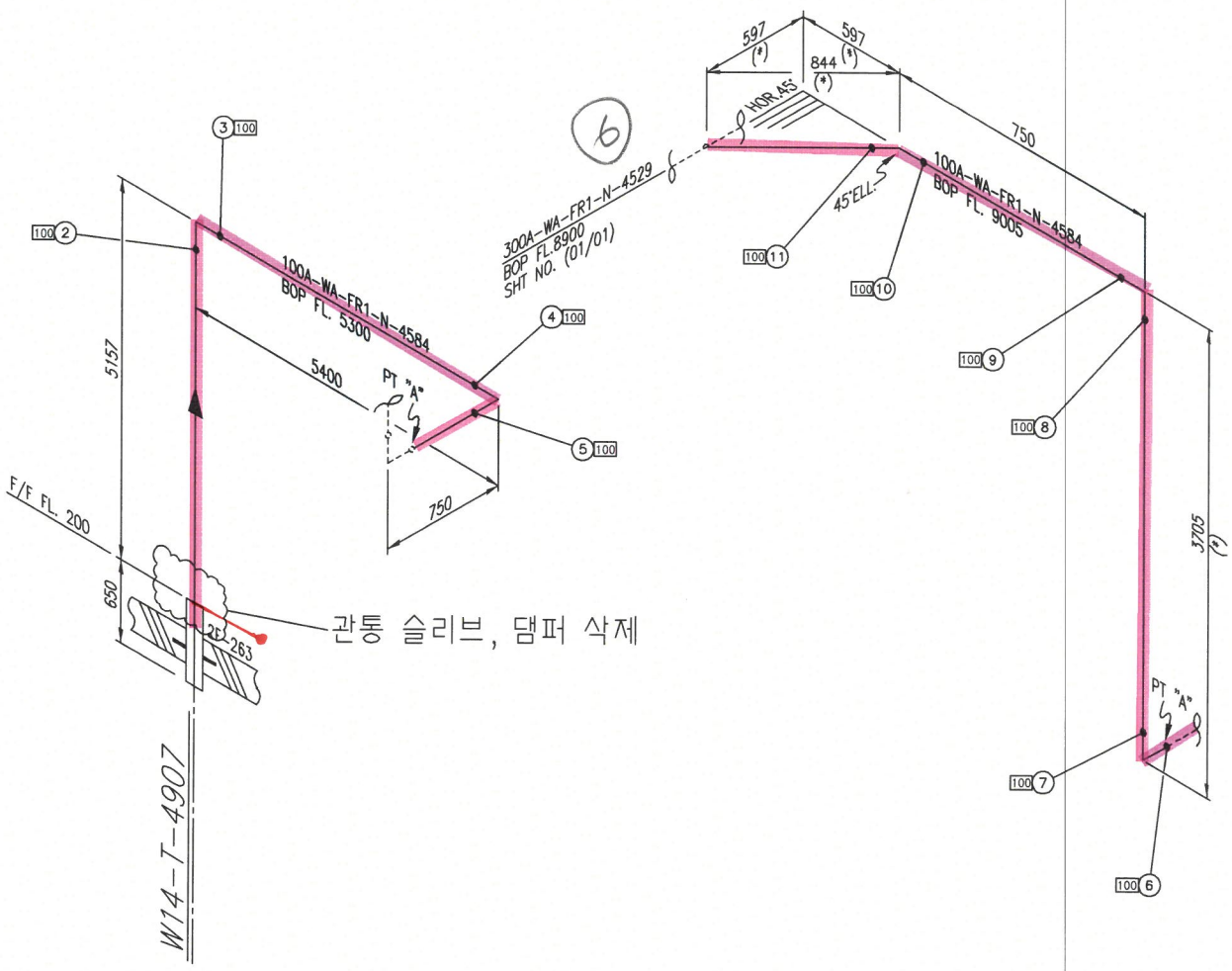
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	1	AMB
OPERATING	ATM	AMB

AREA W14
UNIT 2FA
REF. PLAN DWG. CJ-M15W14-PI-A30-020003-G
REF. P & ID CJ-M15W14-PR-A20-000135

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	24.12.20	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 80A-WA-FR1-N-4647

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	100A	16.6M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
E90	100A	4EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
E45	100A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
DELETE VP	100A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
DELETE F	100A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
DELETE G	100A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
DELETE B	100A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL EPDM LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (24)

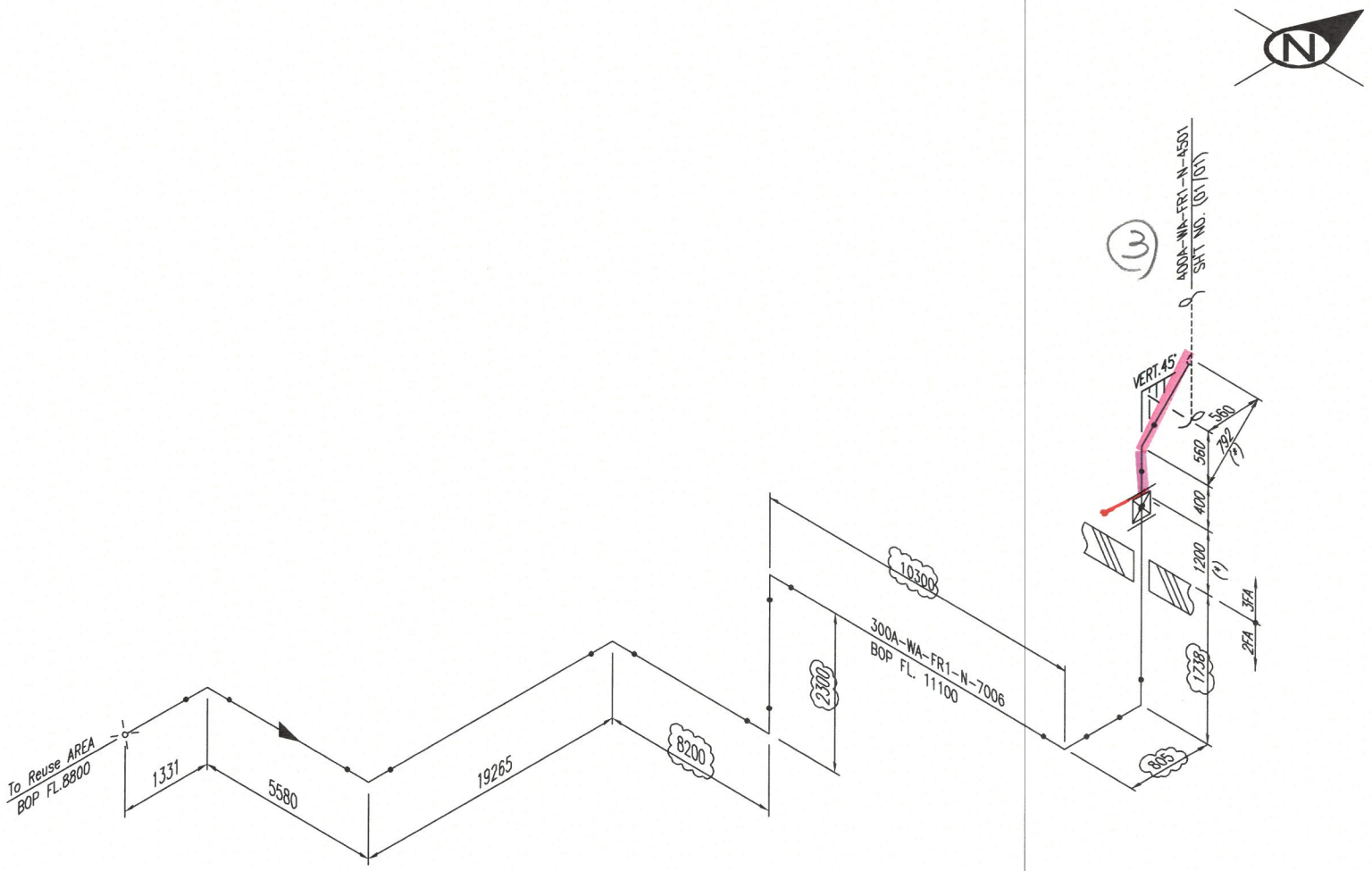
INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	1	AMB
OPERATING	ATM	AMB

AREA W14
UNIT 2FA
REF. PLAN DWG. CJ-M15W14-PI-A30-020003-A
REF. P & ID CJ-M15W14-PR-A20-000163

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY WA
JOB NO. 22330D
LINE NO. 100A-WA-FR1-N-4584
SH'T NO. 01 / 01
REV. 0



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	300A	51.9M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	300A	7EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
E45	300A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
VP	300A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
F	300A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	300A	1EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	300A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90' ELBOW : E90 * 45' ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (25)

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	1	AMB
OPERATING	ATM	AMB

AREA W14
UNIT 2FA
REF. PLAN DWG. CJ-M15W14-PI-A30-020011-G
REF. P & ID CJ-M15W14-PR-A20-000167

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	24.12.20	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY WA
LINE NO. 300A-WA-FR1-N-7006

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

PUNCH LIST

Project No : 22330D

Project Title : M15X Ph3 Project

Package No :

Date :

S/N	Drawing No	Description	Category A B C	Punch Clear Status			
				작성자		확인자	
				Name	Sign	Name	Sign
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Remarks

A : Before Hydro Test

B : Before M/C

C : Before Commissioning